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AIME Geometry Problem Collection

New mathematical restatements of the standard 1983–2026 archive. Contest credit: MAA American Mathematics Competitions. Linked sources supply official wording and diagrams.

1983 AIME Problem 4: The interior corner of a circular notch

Points A and C lie on a circle of radius $\sqrt{50}$. Point B lies inside it, AB=6, BC=2, and $\angle$ABC=90°. Find the squared distance from B to the circle center.

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1983 AIME Problem 11: A square base and an extended ridge

A solid has square base ABCD of side s and upper edge EF of length 2s, parallel to AB. Its faces are ABCD, ADE, BCF, ABFE, and DCEF; AE=DE=BF=CF=s. For $s=6\sqrt{2}$, find its volume.

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1983 AIME Problem 12: A diameter with reversed chord digits

A circle has a two-digit integer diameter AB. A perpendicular chord CD has length obtained by reversing those digits. Their intersection H has positive rational distance from the center O. Find AB.

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1983 AIME Problem 14: Equal chords through an intersection

Two circles of radii 8 and 6 have centers 12 units apart and intersect at P. A line through P meets the first again at Q and the second again at R, with P between Q,R and QP=PR. Find QP^2.

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1983 AIME Problem 15: A unique chord with prescribed bisector

In a circle of radius 5, chord BC has length 6 and bisects chord AD. B lies on the minor arc AD. No other chord from A is bisected by BC. Express the sine of the minor central angle AOB as m/n in lowest terms and find mn.

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1984 AIME Problem 3: Three parallel-line triangles

Point P lies inside triangle ABC. For each side, the two lines through P parallel to the other sides cut off a triangle with vertex P and base on that side. These three triangles have areas 4,9,49. Find the area of ABC.

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1984 AIME Problem 6: Balancing three equal disks

Three disks of radius 3 have centers(14,92),(17,76),(19,84). A line through(17,76) divides their total area equally between its two sides. Find the absolute value of its slope.

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1984 AIME Problem 9: Volume from two face areas

In tetrahedron ABCD, AB=3, areas[ABC]=15 and[ABD]=12, and the dihedral angle between those faces is 30°. Find the volume.

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1985 AIME Problem 2: Rotating a right triangle about either leg

Rotating a right triangle about one leg produces a cone of volume 800π; rotating it about the other produces volume 1920π. Find its hypotenuse.

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1985 AIME Problem 4: A square formed by four sloping lines

In the unit square with vertices(0,0),(1,0),(1,1),(0,1), draw the lines joining these vertices respectively to (1,1-1/n),(1/n,1),(0,1/n),(1-1/n,0), where $n \geq 2$ is an integer. The central square has area 1/1985. Find n.

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1985 AIME Problem 6: Areas around concurrent cevians

Cevians AD,BE,CF meet at an interior point P of triangle ABC, with D∈BC,E∈CA,F∈AB. The areas[APF],[BPF],[BPD],[CPE] are 40,30,35,84 respectively. Find[ABC].

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1985 AIME Problem 9: Three chord lengths and a doubled angle

Chords of lengths 2,3,4 in one circle subtend minor central angles $\alpha,\beta,\alpha+\beta$, with $\alpha+\beta<\pi$. If $\cos\alpha$=m/n in lowest terms, find m+n.

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1985 AIME Problem 11: An ellipse tangent to the axis

An ellipse has foci(9,20) and (49,55) and is tangent to the x-axis. Find the length of its major axis.

Published problem and solutions · MAA AMC.

1985 AIME Problem 15: A hexagonal cut through a cube

Cut three 12×12 squares along segments joining midpoints of adjacent sides, producing three isosceles right triangles and three pentagons. Attach these pieces alternately around a regular hexagon, along their sides of length $6\sqrt{2}$, and fold them into the convex polyhedron specified by this net. Find its volume.

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1986 AIME Problem 9: Equal parallel sections of a triangle

A triangle has side lengths 425,450,510. Through an interior point, draw the full segments inside the triangle parallel to its three sides. All three segments have common length d. Find d.

Published problem and solutions · MAA AMC.

1986 AIME Problem 14: A box from distances to a body diagonal

For a rectangular box, the distances from a body diagonal to the six edges it does not meet take the three values $2\sqrt{5}$,30/$\sqrt{13}$,15/$\sqrt{10}$. Find the box volume.

Published problem and solutions · MAA AMC.

1986 AIME Problem 15: The angle between medians of a right triangle

Triangle ABC is right-angled at C and has hypotenuse 60. Its medians from A and B lie on y=x+3 and y=2x+4. Find its area.

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1987 AIME Problem 2: Maximum distance between two spheres

One point lies on the sphere centered at (-2,-10,5) with radius 19; another lies on the sphere centered at (12,8,-16) with radius 87. Find their largest possible separation.

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1987 AIME Problem 4: A region bounded by absolute values

Find the area enclosed by|x-60|+|y|=|x|/4.

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1987 AIME Problem 6: Four equal regions in a rectangle

In rectangle ABCD, X,Y lie in order on AB and W,Z in order from D to C on DC. Interior points P,Q with PQ‖AB form four equal-area regions via XP,WP,PQ,YQ,ZQ. Suppose XY=YB+BC+CZ=ZW=WD+DA+AX. Given BC=19 and PQ=87, find AB.

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1987 AIME Problem 9: Three equal angles at an interior point

Triangle ABC is right-angled at B. An interior point P satisfies $PA = 10$, $PB = 6$, and $\angle APB = \angle BPC = \angle CPA$. Find PC.

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1987 AIME Problem 15: Two orientations of an inscribed square

Triangle ABC is right-angled at C. One inscribed square has a vertex at C, sides along CA and CB, and its opposite vertex on AB; its area is 441. A second has one side on AB and its other two vertices on CA and CB; its area is 440. Find AC+CB.

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1988 AIME Problem 7: An altitude from a known angle tangent

In triangle ABC, tan∠A=22/7. The altitude from A meets the interior of BC and divides BC into segments 3 and 17. Find the triangle's area.

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1988 AIME Problem 10: Interior diagonals of a polyhedron

A convex polyhedron has 12 square, 8 regular hexagonal and 6 regular octagonal faces, one of each at every vertex. Count vertex-pair segments lying in its interior.

Published problem and solutions · MAA AMC.

1988 AIME Problem 11: Mean lines pass through a centroid

A line is a mean line for complex points $w_1,\ldots,w_5$ if it contains $z_1,\ldots,z_5$ with $\Sigma(z_k\text{-}w_k)=0$. For w=(32+170i,-7+64i,-9+200i,1+27i,-14+43i), find the slope of the mean line whose y-intercept is 3.

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1988 AIME Problem 12: Three equal terminal pieces of cevians

Cevians AD,BE,CF of triangle ABC meet at an interior point P. Suppose PD=PE=PF=3, while AP=a,BP=b,CP=c and a+b+c=43. Find abc.

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1988 AIME Problem 14: Reflecting a rectangular hyperbola

Reflect the curve xy=1 in the line y=2x. Write its equation as $12x^2+bxy+cy^2+d=0$. Find bc.

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1989 AIME Problem 2: Polygons from circle points

Ten points lie on a circle. Count convex polygons with at least three vertices selected from those points.

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1989 AIME Problem 6: The first possible moving rendezvous

A and B are 100 units apart. Allie leaves A at speed 8 along a ray making a 60° angle with AB. Billie leaves B simultaneously at speed 7 along a chosen straight ray. What distance has Allie traveled at their earliest possible meeting?

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1989 AIME Problem 10: Cotangents expressed through area

A triangle has side lengths a,b,c opposite angles α,β,γ, and $a^2+b^2=1989c^2$. Find $\cot\gamma/(\cot\alpha+\cot\beta)$.

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1989 AIME Problem 12: Opposite-edge midpoints in a tetrahedron

A tetrahedron has AB=41, AC=7, AD=18, BC=36, BD=27, and CD=13. If M and N are the midpoints of AB and CD, find MN^2.

Published problem and solutions · MAA AMC.

1989 AIME Problem 15: Area ratios force a right angle

In triangle ABC, cevians AD,BE,CF meet at an interior point P, with $D\in BC, E\in CA, F\in AB$. Given AP=6, BP=9, PD=6, PE=3, and CF=20, find the area of ABC.

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1990 AIME Problem 3: A largest regular-polygon side count

Regular polygons with r and s sides satisfy $r\geq s\geq 3$, and each interior angle of the r-gon is 59/58 times each interior angle of the s-gon. Find the largest possible s.

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1990 AIME Problem 7: An angle bisector from unit vectors

Triangle PQR has P=(−8,5), Q=(−15,−19), and R=(1,−7). Its internal bisector at P has equation $ax+2y+c=0$. Find a+c.

Published problem and solutions · MAA AMC.

1990 AIME Problem 12: Sum of dodecagon chords

For a regular 12-gon of circumradius 12, write the sum of all side and diagonal lengths as $a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}$. Find a+b+c+d.

Published problem and solutions · MAA AMC.

1990 AIME Problem 14: A tetrahedron folded from a rectangle

A $12\sqrt{3}$ by $13\sqrt{3}$ rectangle ABCD has diagonal intersection P. Remove triangle ABP, identify edges AP and BP, and fold along CP and DP to form a tetrahedron. Find its volume.

Published problem and solutions · MAA AMC.

1991 AIME Problem 2: Parallel slices of a rectangle

In a 4 by 3 rectangle ABCD, divide AB and CB into 168 equal parts starting at A and C, and join corresponding interior division points. Repeat on AD and CD, then include diagonal AC. Find the total length of the resulting 335 parallel segments.

Published problem and solutions · MAA AMC.

1991 AIME Problem 11: Twelve disks in the depicted regular ring

In the rotationally symmetric arrangement shown in the original problem, twelve congruent disks form a ring covering a circle of radius 1. Neighboring disks are externally tangent at points of that circle. Their total area is $\pi(a-b\sqrt{c})$, with c squarefree. Find a+b+c.

Published problem and solutions · MAA AMC.

1991 AIME Problem 12: A reflected vertex locates a rhombus center

A rhombus PQRS is inscribed in rectangle ABCD, with P,Q,R,S in the interiors of AB,BC,CD,DA respectively. Given PB=15, BQ=20, PR=30, and QS=40, the rectangle's perimeter is m/n in lowest terms. Find m+n.

Published problem and solutions · MAA AMC.

1991 AIME Problem 14: Three Ptolemy equations for a hexagon

A cyclic hexagon ABCDEF has AB=31 and all its other sides equal to 81. Find AC+AD+AE.

Published problem and solutions · MAA AMC.

1992 AIME Problem 7: A tetrahedron from two face areas

Faces ABC and BCD of a tetrahedron meet at a dihedral angle of 30°. Their areas are 120 and 80, and BC=10. Find the tetrahedron's volume.

Published problem and solutions · MAA AMC.

1992 AIME Problem 9: A base point equidistant from trapezoid legs

Trapezoid ABCD has AB$\|$CD, AB=92, BC=50, CD=19, and AD=70. A circle centered at P on AB is tangent to BC and AD. Write AP=m/n in lowest terms. Find m+n.

Published problem and solutions · MAA AMC.

1992 AIME Problem 10: A square outside two semicircles

Let A consist of nonzero complex z such that the real and imaginary parts of both z/40 and 40/conj(z) lie in [0,1]. Find the integer nearest the area of A.

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1992 AIME Problem 11: The order of two successive reflections

Reflect a line through the origin first in the line making angle $\pi/70$ with the positive x-axis, then in the line making angle $\pi/54$. Starting with y=(19/92)x, find the least positive number of repetitions that restores the original line.

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1992 AIME Problem 13: Maximum area on an Apollonius circle

Triangle ABC has AB=9 and BC:AC=40:41. Find its largest possible area.

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1992 AIME Problem 14: A universal product identity for cevian ratios

Cevians AA′,BB′,CC′ of triangle ABC meet at an interior point O, with A′,B′,C′ on the opposite sides. If AO/OA′+BO/OB′+CO/OC′=92, find the product of these three ratios.

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1993 AIME Problem 2: A forty-day spiral of square distances

A traveler goes east, north, west, south in a repeating four-day cycle. On day n he travels $n^2/2$ miles. How far is he from his starting point after 40 days?

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1993 AIME Problem 10: Uniform vertex face incidences

A convex polyhedron has 32 triangular or pentagonal faces. At each vertex, T triangles and P pentagons meet. Determine 100P+10T+V.

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1993 AIME Problem 12: Undoing random midpoint choices

Inside the triangle with vertices (0,0),(0,420),(560,0), a point undergoes six midpoint moves toward arbitrarily selected vertices and ends at (14,92). Determine the sum of its starting coordinates.

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1993 AIME Problem 13: When a moving line of sight clears a building

Jenny and Kenny walk in the same direction at 1 and 3 feet per second on parallel paths 200 feet apart. A circular building of diameter 100 is centered midway between the paths. When it first blocks their view they are 200 feet apart. If visibility returns after t seconds, find numerator plus denominator of t in lowest terms.

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1993 AIME Problem 14: The least perimeter of an unstuck inscribed rectangle

A rectangle inscribed in a 6 by 8 rectangle has one vertex on each side. Call it unstuck if a nonzero sufficiently small rotation about its center keeps it inside the larger rectangle. The least possible perimeter is $\sqrt{N}$. Find N.

Published problem and solutions · MAA AMC.

1993 AIME Problem 15: Two incircle contacts on an altitude

Triangle ABC has AB=1995, AC=1994, and BC=1993. Its altitude CH meets AB at H. The incircles of ACH and BCH touch CH at R and S. Write RS=m/n in lowest terms and find m+n.

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1994 AIME Problem 2: A square between internally tangent circles

A circle of diameter PQ=10 is internally tangent at P to a circle of radius 20. A square ABCD has A,B on the larger circle, while CD is tangent to the smaller circle at Q and the smaller circle lies outside the square. If AB=m+$\sqrt{\text{n}}$, find m+n.

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1994 AIME Problem 6: Small triangles in three line families

For every integer k from -10 to 10, draw y=k, y=$\sqrt{3}$x+2k and y=$-\sqrt{3}$x+2k. Count equilateral triangles of side $2/\sqrt{3}$ whose edges lie on the lines.

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1994 AIME Problem 7: Lines with only integral circle intersections

Count real pairs (a,b) for which ax+by=1 and $x^2+y^2=50$ have at least one common point and every common point has integral coordinates.

Published problem and solutions · MAA AMC.

1994 AIME Problem 8: An equilateral triangle from two heights

The points (0,0),(a,11),(b,37) form an equilateral triangle. Find ab.

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1994 AIME Problem 10: A prime-cube projection in an integer right triangle

Triangle ABC is right at C and has integer side lengths. The altitude from C meets AB at D with $\text{BD}=29^3$. If cosB=m/n in lowest terms, find m+n.

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1994 AIME Problem 12: The finest square grid within a fence budget

A fenced 24 by 52 meter rectangle is partitioned completely into congruent axis-parallel square plots using at most 1994 meters of additional internal fence. Find the largest possible number of plots.

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1994 AIME Problem 14: Unfolding a beam between two finite mirrors

Two equal-length mirror segments BA and BC meet at angle $\beta=1.994°$. A beam first reflects at C, making angle $\alpha=19.94°$ with BC and then traveling into the angle toward BA. It continues reflecting until it leaves the segments. Count all reflections, including the one at C.

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1994 AIME Problem 15: Fold points as an intersection of diameter disks

For P on triangular paper ABC, fold each vertex onto P, producing the perpendicular-bisector creases restricted to the paper. Call P a fold point if these creases do not intersect. Given AB=36, AC=72, and $\angle$B=90°, the fold-point region has area qπ−r$\sqrt{}$s with s squarefree. Find q+r+s.

Published problem and solutions · MAA AMC.

1995 AIME Problem 1: Five shrinking squares with quarter overlaps

Start with a unit square. Each successive square has half the previous side length and is centered at its outward corner, with parallel sides; at every stage use the corner away from the earlier squares. For the first five squares, write their union area as m/n in lowest terms. Find m−n.

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1995 AIME Problem 4: A common tangent chord of three tangent circles

Circles of radii 3 and 6 are externally tangent and internally tangent to a circle of radius 9. A common external tangent to the smaller circles cuts a chord from the larger circle. Find the square of its length.

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1995 AIME Problem 9: An isosceles triangle with a tripled angle

In isosceles triangle ABC, AB=AC and altitude AM=11. A point D on AM satisfies AD=10 and $\angle$BDC=3$\angle$BAC. If the triangle's perimeter is a+$\sqrt{}$b, find a+b.

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1995 AIME Problem 11: A self-similar cut prism

An integer-sided rectangular prism has sorted sides $a \le b \le c$ with $b = 1995$. A face-parallel cut produces a smaller similar prism. Count possible triples (a, b, c).

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1995 AIME Problem 12: The internal dihedral angle of a square pyramid

A pyramid OABCD has square base ABCD and equal lateral edges OA,OB,OC,OD. Given $\angle$AOB=45°, its internal dihedral angle θ between faces OAB and OBC satisfies cosθ=m+$\sqrt{}$n. Find m+n.

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1995 AIME Problem 14: Two equal chords and an asymmetric region

Two chords of length 78 in a circle of radius 42 meet at a point 18 from the center. Of the four regions they form, consider one bounded by chord segments of unequal lengths. Its area is mπ−n$\sqrt{}$d with d squarefree. Find m+n+d.

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1996 AIME Problem 4: A cube's perspective shadow

A unit cube rests on a horizontal plane. A point light source is directly above one upper vertex, at distance x from that vertex. The shadow area outside the cube's footprint is 48. Find floor(1000x).

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1996 AIME Problem 13: An area ratio from projection onto a median

Triangle ABC has AB=$\sqrt{30}$, AC=$\sqrt{6}$, and BC=$\sqrt{15}$. A point D lies on the line through A and the midpoint of BC, with $\angle$ADB=90°, and segment AD bisects BC. Write [ADB]/[ABC]=m/n in lowest terms. Find m+n.

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1996 AIME Problem 14: Cubes crossed by a space diagonal

A box of side lengths 150,324,375 is partitioned into unit cubes. Count cubes whose interiors meet a long diagonal.

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1996 AIME Problem 15: A parallelogram determined by doubled angles

Diagonals of parallelogram ABCD meet at O. Both $\angle$CAB and $\angle$DBC equal twice $\angle$DBA. If $\angle$ACB=r$\angle$AOB, find floor(1000r).

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1997 AIME Problem 2: Squares among grid rectangles

Let r be the number of rectangles and s the number of squares formed by the nine horizontal and nine vertical lines of a standard 8-by-8 board. Reduce s/r=m/n and find m+n.

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1997 AIME Problem 4: Four mutually tangent circles

Four circles of radii 5,5,8,r are mutually externally tangent. Write r=m/n in lowest terms and find m+n.

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1997 AIME Problem 6: An equilateral attachment and another regular polygon

An equilateral triangle A_1A_2B is attached externally to a regular n-gon $A_1A_2\ldots A_n$. Find the largest n for which A_1,A_n,B can be three consecutive vertices of a regular polygon.

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1997 AIME Problem 7: The midpoint time of a storm crossing

A car moves east at 2/3 mile per minute. A circular storm of radius 51 moves southeast at $\sqrt{2}/2$ mile per minute, initially centered 110 miles north of the car. If the car enters at t_1 and leaves at t_2, find $(t_1+t_2)/2$.

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1997 AIME Problem 13: A wire graph made of unit diagonals

Let f(t)=abs(abs(abs(t)−2)−1). The planar set S is defined by f(x)+f(y)=1. Its total length is a$\sqrt{}$b with b squarefree. Find a+b.

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1997 AIME Problem 14: A short arc probability

Two distinct 1997th roots of unity are chosen uniformly. Find numerator plus denominator for the chance that $|v + w| \geq \sqrt{2 + \sqrt{3}}$.

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1997 AIME Problem 15: A rigorous maximum equilateral triangle in a rectangle

An equilateral triangle lies entirely inside a 10 by 11 rectangle. Its largest possible area is p$\sqrt{}$q−r, where q is squarefree. Find p+q+r.

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1998 AIME Problem 2: Integer points between two slope bounds

Count positive integer pairs with $x \leq 2y \leq 60$ and $y \leq 2x \leq 60$.

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1998 AIME Problem 3: Factoring a graph into four half-lines

The graph y^2+2 xy+40|x|=400 encloses one bounded region. Find its area.

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1998 AIME Problem 6: A geometric mean on a parallelogram transversal

In parallelogram ABCD, extend DA beyond A to P. The line PC meets AB at Q and diagonal DB at R. Given PQ=735 and QR=112, find RC.

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1998 AIME Problem 9: Overlapping coffee breaks

Two people arrive independently and uniformly during one hour and each stays m minutes. Their visits overlap with probability 40%. Write m=a−b$\sqrt{}$c with c squarefree and find a+b+c.

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1998 AIME Problem 10: A sphere tangent to eight spheres on a plane

Eight radius-100 spheres rest on a plane, with centers forming a regular octagon and each tangent to its two neighbors. A ninth sphere rests on the plane and is tangent to all eight. Its radius is a+b$\sqrt{}$c, with c squarefree. Find a+b+c.

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1998 AIME Problem 11: A cube section from its projected strip

Edges AB,BC,CD form a three-edge path in a cube, and AD is a body diagonal. Points P∈AB,Q∈BC,R∈CD satisfy AP=5, PB=15, BQ=15, CR=10. Find the area of the plane section through P,Q,R.

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1998 AIME Problem 12: A golden-ratio configuration in a medial triangle

In equilateral triangle ABC, D,E,F are the midpoints of BC,CA,AB. Points P∈DE,Q∈EF,R∈FD satisfy P∈CQ, Q∈AR, and R∈BP. If [ABC]/[PQR]=a+b$\sqrt{c}$ with c squarefree, find $a^2+b^2+c^2$.

Published problem and solutions · MAA AMC.

1999 AIME Problem 2: A congruent cut through a parallelogram center

A line through the origin cuts the parallelogram with vertices (10,45),(10,114),(28,153),(28,84) into two congruent polygons. If its slope is m/n in lowest terms, find m+n.

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1999 AIME Problem 4: The intersection of two concentric unit squares

Two unit squares have the same center and intersect in an octagon. One side of the octagon has length 43/99. If its area is m/n in lowest terms, find m+n.

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1999 AIME Problem 6: Square roots turn a quadrilateral into an annular sector

Apply (x,y)↦($\sqrt{x}$,$\sqrt{y}$) to the region enclosed by (900,300),(1800,600),(600,1800),(300,900). If its image has area k, find floor(k).

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1999 AIME Problem 8: Exactly two thresholds on a simplex

On the nonnegative triangle $x + y + z = 1$, take points satisfying exactly two of $x \geq 1/2, y \geq 1/3$, $z \geq 1/6$. If their area fraction is m/n reduced, find $m + n$.

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1999 AIME Problem 9: An equidistant complex multiplier

For positive real a,b, the map f(z)=(a+bi)z sends every complex z to a point equidistant from z and 0. Given |a+bi|=8 and b^2=m/n in lowest terms, find m+n.

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1999 AIME Problem 12: A triangle from inradius and two tangent lengths

The incircle of triangle ABC has radius 21 and touches AB at P, with AP=23 and PB=27. Find the triangle's perimeter.

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1999 AIME Problem 14: The common angle at a Brocard point

Inside a triangle with AB=13, BC=14, CA=15, a point P satisfies $\angle$PAB=$\angle$PBC=$\angle$PCA=θ. Write tanθ=m/n in lowest terms and find m+n.

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1999 AIME Problem 15: Folding a medial triangle into a tetrahedron

A paper triangle has vertices (0,0),(34,0),(16,24). Fold along the three sides of its medial triangle so the original vertices meet at a tetrahedron apex. Find the volume.

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2000 AIME I Problem 2: A reflected pentagon

Let $A = (u, v)$, where u, v are integers with $0 < v < u$. Obtain B, C, D, E successively by reflecting A across $y = x$, B across the y-axis, C across the x-axis, and D across the y-axis. If pentagon $ABCDE$ has area 451, find $u + v$.

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2000 AIME I Problem 4: Nine-square dissection as a linear system

A rectangle is tiled by nine squares with side lengths $a_1 < \cdots < a_9$. Their edge relations are $a_3 = a_1 + a_2$, $a_4 = a_1 + a_3$, $a_5 = a_3 + a_4$, $a_6 = a_4 + a_5$, $a_7 = a_2 + a_3 + a_5$, $a_8 = a_2 + a_7$, $a_9 = a_1 + a_4 + a_6$, and $a_6 + a_9 = a_7 + a_8$. Its width is $a_8 + a_9$ and height is $a_6 + a_9$. If these dimensions are coprime integers, find its perimeter.

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2000 AIME I Problem 8: Turning a cone of liquid upside down

A right circular cone of height 12 and base radius 5 contains liquid to depth 9 when its vertex points down. After it is turned vertex-up, the liquid depth is $m - n\sqrt[3]{p}$, with positive integers m, n, p and cube-free p. Find $m + n + p$.

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2000 AIME I Problem 13: Fast highways and a slower prairie

Two perpendicular highways cross at the fire station. A truck travels at 50 miles per hour on either highway and 14 miles per hour elsewhere. The area it can reach in six minutes is m/n square miles in lowest terms. Find $m + n$.

Published problem and solutions · MAA AMC.

2000 AIME I Problem 14: An isosceles chain of equal segments

In triangle ABC, $AB = AC$. Points P on AC and Q on AB satisfy $AP = PQ = QB = BC$. Find $\lfloor 1000\angle ACB/\angle APQ \rfloor$.

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2000 AIME II Problem 6: A parallel area bisector

The bases of a trapezoid differ by 100. Its midline divides its area in the ratio $2:3$. A parallel segment of length x bisects its area. Find $\lfloor x^2/100 \rfloor$.

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2000 AIME II Problem 8: A right trapezoid with perpendicular diagonals

In trapezoid $ABCD$, $AB \parallel CD$, BC is perpendicular to the bases, and $AC \perp BD$. Given $AB = \sqrt{11}$ and $AD = \sqrt{1001}$, determine BC^2.

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2000 AIME II Problem 10: Radius from four tangent lengths

A circle inscribed in convex quadrilateral $ABCD$ touches AB at P and CD at Q. If $AP = 19$, $PB = 26$, $CQ = 37$, and $QD = 23$, find the square of its radius.

Published problem and solutions · MAA AMC.

2000 AIME II Problem 11: Integral equal legs restrict base directions

An isosceles trapezoid $ABCD$ has integer-coordinate vertices, $A = (20, 100)$, $D = (21, 107)$, no horizontal or vertical side, and only $AB \parallel CD$. The sum of the absolute values of all possible slopes of AB is m/n in lowest terms. Find $m + n$.

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2000 AIME II Problem 12: A triangular section of a sphere

Three points A, B, C on a sphere of radius 20 satisfy $AB = 13$, $BC = 14$, $CA = 15$. The distance from the center to triangle ABC is $m\sqrt{n}/k$, with m, k coprime positive integers and square-free n. Find $m + n + k$.

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2001 AIME I Problem 4: An angle bisector creates an isosceles triangle

In triangle ABC, $\angle A = 60^\circ$, $\angle B = 45^\circ$. The internal bisector AT meets BC and has length 24. Write the area as $a + b\sqrt{c}$, with positive integers and square-free c, and find $a + b + c$.

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2001 AIME I Problem 5: Equilateral triangle in an ellipse

An equilateral triangle inscribed in $x^2 + 4y^2 = 4$ has vertex $(0, 1)$ and an altitude on the y-axis. If its squared side length is reduced m/n, find $m + n$.

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2001 AIME I Problem 7: A parallel through the incenter

In triangle ABC, $AB = 21$, $AC = 22$, and $BC = 20$. The line through the incenter parallel to BC meets AB, AC at D, E. Write $DE = m/n$ in lowest terms and find $m + n$.

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2001 AIME I Problem 9: Area from symmetric side fractions

In triangle ABC with sides $13, 15, 17$, points $D \in AB, E \in BC, F \in CA$ satisfy $AD/AB = p$, $BE/BC = q$, $CF/CA = r$, where $p, q, r > 0$, $p + q + r = 2/3$, and $p^2 + q^2 + r^2 = 2/5$. If $[DEF]/[ABC] = m/n$ in lowest terms, find $m + n$.

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2001 AIME I Problem 10: Parity classes of lattice points

Choose two distinct points uniformly from $\{0, 1, 2\} \times \{0, 1, 2, 3\} \times \{0, 1, 2, 3, 4\}$. Their midpoint belongs to the same set with probability m/n in lowest terms. Find $m + n$.

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2001 AIME I Problem 12: Insphere of a coordinate tetrahedron

A sphere is inscribed in the tetrahedron with vertices $(0, 0, 0), (6, 0, 0), (0, 4, 0), (0, 0, 2)$. Its radius is reduced m/n. Find $m + n$.

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2001 AIME I Problem 13: Three successive equal arcs

In a circle, a chord subtending an arc of d° has length 22, and the chord of a $2d^\circ$ arc is 20 longer than the chord of a $3d^\circ$ arc, where $0 < d < 120$. If the latter chord is $-m + \sqrt{n}$, find $m + n$.

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2001 AIME I Problem 15: Hamilton cycles in a cube complement

Label the eight faces of a regular octahedron randomly by $1, \ldots, 8$. No consecutive labels, including 8, 1, may share an edge. Express the probability as reduced m/n and find $m + n$.

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2001 AIME II Problem 4: A prescribed midpoint between two lines

Points P, Q lie on $8y = 15x$ and $10y = 3x$, respectively, and have midpoint $(8, 6)$. If $PQ = m/n$ in lowest terms, find $m + n$.

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2001 AIME II Problem 6: A small square between a chord and a circle

A unit-area square $ABCD$ is inscribed in a circle. Another square $EFGH$ has E, F on side CD and G, H on the circle. Its area is reduced $m/n < 1$. Find $10n + m$.

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2001 AIME II Problem 7: Incircle cuts in a right triangle

In right triangle PQR with $PQ = 90$, $PR = 120$, $QR = 150$, the non-leg tangent to its incircle perpendicular to PR cuts off triangle RST at R; the analogous tangent perpendicular to PQ cuts off triangle QUV at Q. If the distance between the two new incenters is $\sqrt{10n}$, find n.

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2001 AIME II Problem 12: Recursive tetrahedral growth

Begin with a unit-volume regular tetrahedron. At each stage replace the middle quarter-triangle of every triangular face by the three exposed faces of an outward regular tetrahedron. Find numerator plus denominator of the volume after three stages.

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2001 AIME II Problem 13: An angle bisector reveals a similarity

In convex quadrilateral $ABCD$, $\angle BAD = \angle ADC$ and $\angle ABD = \angle BCD$. Given $AB = 8$, $BD = 10$, $BC = 6$, write $CD = m/n$ in lowest terms and find $m + n$.

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2001 AIME II Problem 15: A triangular tunnel along a cube diagonal

In the cube $[0, 8]^3$, drill the triangular prism parallel to $(1, 1, 1)$ whose cross-section has vertices $(2, 0, 0)$, $(0, 2, 0), (0, 0, 2)$. Include the three tunnel walls in the remaining surface area. If that area is $m + n\sqrt{p}$, with positive integers and square-free p, find $m + n + p$.

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2002 AIME I Problem 2: Three staggered rows of circles

Twenty congruent circles are packed in three horizontal rows of $7, 6, 7$. Adjacent circles in each row are tangent; each middle-row circle touches the two neighboring circles in each outer row. The enclosing rectangle is tangent to the extreme circles. Its longer-to-shorter side ratio is $(\sqrt{p} - q)/2$. Find $p + q$.

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2002 AIME I Problem 5: Squares through dodecagon vertices

How many squares in the plane have at least two vertices among the vertices of a fixed regular dodecagon?

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2002 AIME I Problem 10: Two area cuts along an angle bisector

Triangle ABC is right at B, and its angle bisector AD meets BC. Points $E \in AB, F \in AC$ satisfy $AE = 3, EB = 9, AF = 10, FC = 27$. Let $G = AD \cap EF$. Find the integer nearest to the area of quadrilateral $DCFG$.

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2002 AIME I Problem 11: Unfolding a reflected ray in a cube

A ray starts at a vertex of a cube of side 12. Its first reflection is on the opposite face at offsets 7 and 5 from the two adjacent edges through the corresponding vertex. It reflects specularly until reaching another cube vertex. If the path length is $m\sqrt{n}$ with square-free n, find $m+n$.

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2002 AIME I Problem 13: A centroid and a secant through a midpoint

Triangle ABC has medians $AD = 18$ and $CE = 27$, with $AB = 24$. Extend CE through E to its second intersection F with the circumcircle. If $[AFB] = m\sqrt{n}$ with square-free n, find $m+n$.

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2002 AIME I Problem 15: Coplanar faces determine a spatial edge

A polyhedron has faces $ABCD, ABFG, CDE, ADEG, BCEF, EFG$. The first is a square of side 12; $ABFG$ is an isosceles trapezoid with $AB \parallel GF$, $AG = BF = 8$, $GF = 6$; and $CE = DE = 14$. The distance from E to square face $ABCD$ is 12. If $EG^2 = p - q\sqrt{r}$ with positive integers and square-free r, find $p+q+r$.

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2002 AIME II Problem 2: An equilateral triangle of cube vertices

Three vertices of a cube are $(7, 12, 10), (8, 8, 1), (11, 3, 9)$. Find its surface area.

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2002 AIME II Problem 4: Garden inside a hexagonal ring

Unit-side regular hexagonal tiles lie in a hexagonal grid. A single-cell-wide outer hexagonal ring has 202 tiles along each straight side, including its corner tiles. Find the enclosed area excluding the ring, expressed as $m\sqrt{3}/2$, modulo 1000 in the coefficient m.

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2002 AIME II Problem 13: An affine height from intersecting cevians

In triangle ABC, $CD = 2, DB = 5$ on BC, and $CE = 1, EA = 3$ on CA, with $AB = 8$. Let $P = AD \cap BE$. Through P, lines parallel to CA, CB meet AB at Q, R, respectively. If $[PQR]/[ABC] = m/n$ in lowest terms, find $m+n$.

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2002 AIME II Problem 14: A circle centered on a leg of a right triangle

Right triangle APM has right angle at A and perimeter 152. A circle of radius 19, centered at $O \in AP$, is tangent to AM and PM. Write $OP = m/n$ in lowest terms and find $m+n$.

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2002 AIME II Problem 15: Two tangent circles through one point

Two circles meet in two points, including $(9, 6)$. Both are tangent to the x-axis and to $y = mx$, where $m > 0$, and their radii have product 68. Write $m = a\sqrt{b}/c$ in lowest radical form and find $a + b + c$.

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2003 AIME I Problem 2: Alternating concentric annuli

Concentric circles have radii $1, 2, \ldots, 100$. Color the central disk red and alternate red and green on successive annuli. If the green area divided by the radius-100 disk area is reduced m/n, find $m + n$.

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2003 AIME I Problem 5: The unit neighborhood of a box

Find the volume of all points whose Euclidean distance from a 3 by 4 by 5 rectangular box is at most 1. Write it as $(m + n\pi)/p$ with positive integers and $\gcd(n, p) = 1$, and find $m + n + p$.

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2003 AIME I Problem 6: Triangle types in a cube

Sum the areas of all triangles with vertices among the eight vertices of a unit cube. Write the sum as $m + \sqrt{n} + \sqrt{p}$ with integer parameters and find $m + n + p$.

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2003 AIME I Problem 7: Integral distances in an isosceles triangle

Points A, B, C are collinear in that order with $AB = 9, BC = 21$. A noncollinear point D satisfies $AD = CD$, and both AD and BD are integers. Find the sum of all possible perimeters of triangle ACD.

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2003 AIME I Problem 10: A hidden isosceles triangle from complementary angles

In isosceles triangle ABC, $AC = BC$ and $\angle ACB = 106^\circ$. Interior point M satisfies $\angle MAC = 7^\circ$ and $\angle MCA = 23^\circ$. Find $\angle CMB$ in degrees.

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2003 AIME I Problem 11: Triangle inequalities as interval probability

Choose x uniformly from $(0, 90^\circ)$. The probability that $\sin^2 x, \cos^2 x, \sin x \cos x$ fail to form a triangle is d/n, where d is the degree measure of $\arctan m$ and $m + n < 1000$. Find $m + n$.

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2003 AIME I Problem 12: Equal opposite angles cancel two cosine laws

Convex quadrilateral $ABCD$ has $\angle A = \angle C$, $AB = CD = 180$, $AD \neq BC$, and perimeter 640. Find $\lfloor 1000 \cos A \rfloor$.

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2003 AIME I Problem 15: Median crossing a perpendicular to an angle bisector

Triangle ABC has $AB = 360, BC = 507, CA = 780$. Let M be the midpoint of CA and let the internal angle bisector BD meet CA at D. Through D draw the perpendicular to BD, meeting BC at F and BM at E. Write $DE/EF = m/n$ in lowest terms and find $m + n$.

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2003 AIME II Problem 4: The tetrahedron of face centroids

The centers of the four faces of a regular tetrahedron form another tetrahedron. Write the smaller-to-larger volume ratio as reduced m/n and find $m + n$.

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2003 AIME II Problem 5: A tangent wedge of a cylinder

A cylindrical log has diameter 12. A plane perpendicular to its axis and another plane making 45° with it cut off a wedge. Their intersection line touches the log at exactly one point. If the wedge volume is $n\pi$, find n.

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2003 AIME II Problem 6: A triangle united with its centroid reflection

Triangle ABC has side lengths $13, 14, 15$ and centroid G. Reflect all three vertices through G to obtain triangle $A'B'C'$. Find the area of the union of the two triangular regions.

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2003 AIME II Problem 7: A rhombus from two circumradii

In rhombus ABCD, triangles ABD and ACD have circumradii 25/2 and 25, respectively. Find the rhombus area.

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2003 AIME II Problem 11: Area using a perpendicular bisector as the base

Right triangle ABC has $AC = 7$, $BC = 24$, and right angle C. Let M be the midpoint of AB, and let D lie on C's side of AB with $AD = BD = 15$. Write $[CDM] = m\sqrt{n}/p$ in lowest radical form and find $m + n + p$.

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2003 AIME II Problem 14: An equilateral hexagon with six prescribed heights

A convex equilateral hexagon ABCDEF has $A = (0, 0)$, $B = (b, 2)$, $\angle FAB = 120°$, and opposite sides parallel. Its vertex y-coordinates are exactly $0, 2, 4, 6, 8, 10$. Write its area as $m\sqrt{n}$ with square-free n, and find $m + n$.

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2004 AIME I Problem 3: Pairs of polyhedron vertices

A convex polyhedron has 26 vertices, 60 edges, 24 triangular faces and 12 quadrilateral faces. Count pairs of vertices whose segment is neither an edge nor contained in a face.

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2004 AIME I Problem 4: Midpoints of sliding fixed-length chords

In a square of side 2, consider every length-2 segment with endpoints on adjacent sides. Its midpoints trace a closed boundary. If the enclosed area rounded to two decimal places is k, find $100k$.

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2004 AIME I Problem 8: Regular star polygons

Count similarity classes of connected regular 1000-vertex star polygons with equal edge lengths and equal counterclockwise turning angles less than 180°, requiring a genuine edge crossing.

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2004 AIME I Problem 9: Matching triangle-and-trapezoid dissections

A 3-4-5 triangle and a 6 by 7 rectangle are each cut by one segment into a triangle U and a trapezoid V. The two U pieces are similar, and the two V pieces are similar. Find m+n if the smallest possible area of the triangle piece in the original triangle is reduced m/n.

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2004 AIME I Problem 10: A random disk avoids a diagonal

A unit-radius disk has its center uniformly distributed among positions where it fits in a 15×36 rectangle. Find numerator plus denominator of the probability that it misses a fixed diagonal.

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2004 AIME I Problem 11: Equal painted-area and volume ratios

A cone of height 4 and radius 3 is painted on its entire exterior, including its base. A parallel cut separates a smaller cone C and a frustum F. Their painted-area ratio C:F equals their volume ratio C:F, namely reduced m/n. Find m+n.

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2004 AIME I Problem 12: Area selected by even logarithmic floors

Let S consist of $(x, y) \in (0, 1]^2$ for which both $\lfloor \log_2(1/x) \rfloor$ and $\lfloor \log_5(1/y) \rfloor$ are even. If S has area reduced m/n, find m+n.

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2004 AIME I Problem 14: A taut rope partly wrapped around a cylinder

A rope of length 20 runs taut from ground level on a cylindrical tower of radius 8 to an endpoint 4 above ground and 4 from the tower surface. Part of the rope lies against the tower. Its contact length is $(a - \sqrt{b})/c$, where a,b,c are positive integers and c is prime. Find a+b+c.

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2004 AIME II Problem 1: A chord halfway along a radius

A chord is perpendicular to a radius at its midpoint. The ratio of the larger circular region to the smaller is $(a\pi + b\sqrt{c})/(d\pi - e\sqrt{f})$ with positive integers, square-free c,f, and $\gcd(a, e) = 1$. Find $abcdef$ mod 1000.

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2004 AIME II Problem 3: Hidden cubes

A rectangular block made of unit cubes has exactly 231 cubes invisible when three mutually adjacent faces are viewed. Find the least possible total number of cubes.

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2004 AIME II Problem 7: A fold as a perpendicular bisector

Rectangle ABCD is folded so B maps to B' on AD. The crease meets AB at E and CD at F, with $AE = 8, EB = 17, CF = 3$. If its perimeter is reduced m/n, find m+n.

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2004 AIME II Problem 11: A shortest path on a cone

A right circular cone has radius 600 and height $200\sqrt{7}$. Two points on opposite generators are at distances 125 and $375\sqrt{2}$ from the vertex. Find the shortest distance along the cone surface between them.

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2004 AIME II Problem 12: A circle between four vertex circles

Isosceles trapezoid ABCD has $AB = 6, CD = 4, AD = BC = 5$. Circles centered at A,B have radius 3, and those at C,D have radius 2. A circle contained in the trapezoid is tangent to all four. Write its radius as $(-k + m\sqrt{n})/p$ with square-free n and $\gcd(k, p) = 1$, and find $k + m + n + p$.

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2004 AIME II Problem 13: A pentagon built from three parallel directions

Convex pentagon ABCDE satisfies $AB \parallel CE$, $BC \parallel AD$, $AC \parallel DE$, $\angle ABC = 120^\circ$, $AB = 3$, $BC = 5$, and $DE = 15$. If $[ABC]/[EBD] = m/n$ in lowest terms, find m+n.

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2005 AIME I Problem 1: A ring of six tangent circles

Six congruent circles form a closed ring, each externally tangent to its two neighbors and internally tangent to a circle of radius 30. If K is the area inside the large circle but outside all six small circles, find $\lfloor K \rfloor$.

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2005 AIME I Problem 7: Two sixty-degree base angles

Quadrilateral ABCD has $BC = 8$, $CD = 12$, $AD = 10$, and $\angle A = \angle B = 60°$. If $AB = p + \sqrt{q}$ with positive integers p,q, find p+q.

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2005 AIME I Problem 9: Local exposure types in a random cube

Each of 27 unit cubes has four painted faces, with its two unpainted faces adjacent. Orient them independently uniformly in a $3 \times 3 \times 3$ cube. If the probability that the outer surface is painted is $p^a/(q^b r^c)$ in prime form, find $a + b + c + p + q + r$.

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2005 AIME I Problem 10: Move along a median to prescribe area

Triangle ABC has area 70, with $B = (12, 19)$ and $C = (23, 20)$. Its median from $A = (p, q)$ to BC has slope -5. Find the largest p+q.

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2005 AIME I Problem 11: Largest semicircle in a square

A semicircular region lies inside a square of side 8. Its largest possible diameter is $m - \sqrt{n}$ for positive integers m,n. Find m+n.

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2005 AIME I Problem 14: Constructing a square by perpendicular normal coordinates

A unique square has the points $A = (0, 12)$, $B = (10, 9)$, $C = (8, 0)$, $D = (-4, 7)$ on four different sides. If its area is K, find the remainder of 10K modulo 1000.

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2005 AIME I Problem 15: An incircle trisecting a median

Triangle ABC has $BC = 20$. Its incircle meets median AD at the two trisection points. If its area is $m\sqrt{n}$ with square-free n, find m+n.

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2005 AIME II Problem 8: A common tangent chord in an enclosing circle

Circles of radii 4 and 10 are externally tangent and internally tangent to a third circle, with all centers collinear. A common external tangent of the smaller circles cuts a chord from the third. Write its length as $m\sqrt{n}/p$ in lowest radical form and find m+n+p.

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2005 AIME II Problem 10: An octahedron and its face-center cube

For a regular octahedron O, let C be the cube whose vertices are its face centers. Write $\mathrm{Vol}(O)/\mathrm{Vol}(C) = m/n$ in lowest terms and find m+n.

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2005 AIME II Problem 12: A forty-five-degree view of a segment

In square ABCD of side 900 and center O, points E,F lie on AB in that order, with $AE < BF$, $EF = 400$, and $\angle EOF = 45^\circ$. If $BF = p + q\sqrt{r}$ with square-free r, find p+q+r.

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2005 AIME II Problem 14: Isogonal cevians multiply side ratios

Triangle ABC has $AB = 13$, $BC = 15$, $CA = 14$. Point D lies on BC with $CD = 6$. Point E lies on BC and satisfies $\angle BAE = \angle CAD$. If $BE = p/q$ in lowest terms, find q.

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2005 AIME II Problem 15: Tangent-circle centers form an ellipse

Circles w_1, w_2 have equations $x^2 + y^2 + 10x - 24y - 87 = 0$ and $x^2 + y^2 - 10x - 24y + 153 = 0$. Let m be the least positive slope of a line through the origin containing the center of a circle internally tangent to w_1 and externally tangent to w_2. If $m^2 = p/q$ in lowest terms, find p+q.

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2006 AIME I Problem 1: Two successive right triangles

Quadrilateral $ABCD$ has $AB = 18$, $BC = 21$, $CD = 14$, $AB \perp BC$, and $AC \perp CD$. Find its perimeter.

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2006 AIME I Problem 7: Alternating strips in an angle

An angle has its vertex between parallel lines L_0, L_1 in an equally spaced family. Both rays cross L_1, $L_2, \ldots$. Let A be its triangular part before L_1 and let B, C, D be its portions between L_2, L_3, between L_4, L_5, and between L_6, L_7, respectively. If $[C]/[B] = 11/5$, find $[D]/[A]$.

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2006 AIME I Problem 8: Possible areas of the central rhombus

Four congruent rhombi surround a fifth rhombus. All five have a common side length s; each outer rhombus has acute angle θ, and the central rhombus has an angle 2θ, where $0 < \theta < 90^\circ$. Each outer area is $\sqrt{2006}$. How many positive integer areas can the central rhombus have?

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2006 AIME I Problem 10: A line bisecting eight disks

Eight radius-1/2 disks have centers $(i + 1/2, j + 1/2)$ for $i, j \in \{0, 1, 2\}$, omitting $(5/2, 5/2)$. A line of slope 3 bisects their total area. Write it as $ax = by + c$ with positive coprime integers a, b, c, and find $a^2 + b^2 + c^2$.

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2006 AIME I Problem 14: Height of a shortened tripod

Three rigid tripod legs of length 5 meet pairwise at equal angles, placing their common top 4 above the ground. One leg is shortened to length 4, without changing the angles between legs. The new height is $m/\sqrt{n}$ in lowest squarefree-radical form. Find $\lfloor m+\sqrt{n}\rfloor$.

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2006 AIME II Problem 1: An equilateral hexagon with two right angles

A convex hexagon has six equal sides. Two opposite angles are 90°, and the other four angles are equal. Its area is $2116(1+\sqrt{2})$. Find its side length.

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2006 AIME II Problem 2: Logarithms as triangle sides

For how many positive integers n do $\log_{10} 12$, $\log_{10} 75$, and $\log_{10} n$ form the sides of a nondegenerate triangle?

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2006 AIME II Problem 6: An axis-parallel square in a right triangle

In unit square $ABCD$, points $E \in BC$ and $F \in CD$ make AEF equilateral. A smaller square lies inside $ABCD$, has vertex B, parallel sides, and its opposite vertex on AE. Its side is $(a-\sqrt{b})/c$ in reduced form with squarefree b. Find $a+b+c$.

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2006 AIME II Problem 9: Intersecting internal common tangents

Circles with centers $(0,0), (12,0), (24,0)$ have radii $1, 2, 4$. The first two have a common internal tangent of positive slope; the last two have one of negative slope. Their intersection has x-coordinate $p-q\sqrt{r}$ in reduced squarefree form. Find $p+q+r$.

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2006 AIME II Problem 12: A cyclic triangle and a parallelogram diagonal

An equilateral triangle ABC has circumradius 2. Points D beyond B on ray AB and E beyond C on ray AC satisfy $AD = 13$, $AE = 11$. Complete parallelogram $ADFE$. Let $G \neq A$ be the second intersection of AF with the circumcircle. Express $[BCG] = p\sqrt{q}/r$ in reduced squarefree form and find $p+q+r$.

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2006 AIME II Problem 15: A radical system encoding triangle altitudes

Real numbers satisfy $x = \sqrt{y^2-1/16}+\sqrt{z^2-1/16}$, $y = \sqrt{z^2-1/25}+\sqrt{x^2-1/25}$, and $z = \sqrt{x^2-1/36}+\sqrt{y^2-1/36}$. If $x+y+z = m/\sqrt{n}$ with squarefree n and positive integers m, n, find $m+n$.

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2007 AIME I Problem 9: Two equal exterior tangent circles

Right triangle ABC has $CA = 30$, $CB = 16$. Outside the triangle, two equal-radius circles touch the hypotenuse and, respectively, the extensions of CA beyond A and CB beyond B. They also touch each other externally. If the common radius is p/q in lowest terms, find $p + q$.

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2007 AIME I Problem 12: Overlap of two rotated isosceles triangles

Let $A = (0, 0)$, $B = (20, 0)$, and let C lie in the first quadrant with $AC = BC$ and $\angle BAC = 75^\circ$. Rotate the triangle counterclockwise until C reaches the positive vertical axis. If the common area is $p\sqrt{2} + q\sqrt{3} + r\sqrt{6} + s$, find $(p - q + r - s)/2$.

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2007 AIME I Problem 13: A pentagonal section of a square pyramid

A square pyramid $ABCDE$ has all eight edges of length 4, with base $ABCD$. A plane passes through the midpoints of AE, BC, and CD. Its section has area $\sqrt{p}$. Find p.

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2007 AIME I Problem 15: An inscribed sixty-degree triangle

In equilateral triangle ABC, points $F \in AB$, $D \in BC$, and $E \in CA$ satisfy $AF = 5$, $CD = 2$, $\angle DEF = 60^\circ$, and $[DEF] = 14\sqrt{3}$. The two possible side lengths are $p \pm q\sqrt{r}$ with rational p, q and squarefree integer r. Find r.

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2007 AIME II Problem 3: Two exterior right triangles on a square

Square $ABCD$ has side 13. Exterior points E, F satisfy $AE = CF = 12$ and $BE = DF = 5$. Find EF^2.

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2007 AIME II Problem 5: Grid cells below a rational line

Count unit grid squares whose interiors lie in the first quadrant and below $9x + 223y = 2007$.

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2007 AIME II Problem 9: Two incircle contact points on one side

Rectangle $ABCD$ has $AB = 63$ and $BC = 448$. Points $E \in AD$ and $F \in BC$ satisfy $AE = CF = 84$. The incircles of BEF and DEF touch EF at P, Q, respectively. Find PQ.

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2007 AIME II Problem 11: Rolling over a smaller cylinder

A radius-72 cylinder rolls along a plane, over a fixed parallel radius-24 cylinder, and back onto the plane without slipping. After exactly one complete rotation, its horizontal displacement is $a\pi + b\sqrt{c}$, with integer coefficients and squarefree c. Find $a + b + c$.

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2007 AIME II Problem 15: Four equal circles in a triangle

Inside a triangle of side lengths $13, 14, 15$, three equal circles are tangent to the respective pairs of sides at the three vertices. A fourth circle of the same radius is externally tangent to all three. If the radius is m/n in lowest terms, find $m + n$.

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2008 AIME I Problem 2: A triangle overlapping a square

An isosceles triangle has as its base one side of a side-10 square. Their overlap has area 80. Find the altitude of the triangle to that base.

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2008 AIME I Problem 5: A cone rolling around its vertex

A right circular cone of base radius r and height h rolls on a plane without slipping, keeping its vertex fixed. During its first full circuit around the vertex it makes 17 rotations. Write $h/r = m\sqrt{n}$ with squarefree n, and find $m + n$.

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2008 AIME I Problem 10: A trapezoid forced by equality bounds

An isosceles trapezoid $ABCD$ has longer base $AD \parallel BC$, base angles 60°, and diagonals $10\sqrt{21}$. A point E satisfies $EA = 10\sqrt{7}$, $ED = 30\sqrt{7}$. If F is the perpendicular projection of C onto AD, write $EF = m\sqrt{n}$ in squarefree form and find $m + n$.

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2008 AIME I Problem 14: Maximizing a tangent projection distance

Circle ω has diameter $AB = 18$. A point C lies beyond A on ray BA, and CT is tangent to the circle. Let P be the perpendicular projection of A onto CT. If m is the largest possible BP, find m^2.

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2008 AIME I Problem 15: The height of a folded paper tray

At each corner of a side-100 square, mark points distance $\sqrt{17}$ along the adjacent edges. Cut from both marks to a common interior point on the corner diagonal, with angle 60° between the cuts, removing the corner wedge. Fold along the lines joining neighboring cut-intersection points and join each pair of cut edges to form a tray. Its height is $\sqrt[n]{m}$ in reduced form, $m < 1000$. Find $m + n$.

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2008 AIME II Problem 3: Ten slices from a rectangular block

A $10 \times 13 \times 14$ rectangular solid loses ten slices, each one unit thick and parallel to a face of the remaining block. What is the largest possible remaining volume?

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2008 AIME II Problem 5: Midpoints of bases with complementary base angles

Trapezoid $ABCD$ has $BC \parallel AD$, $BC = 1000$, $AD = 2008$, and base angles $\angle A = 37^\circ$, $\angle D = 53^\circ$. If M, N are the midpoints of BC, AD, find MN.

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2008 AIME II Problem 9: An affine rotation with period eight

Start at $(5, 0)$. A move rotates the point counterclockwise by 45° about the origin and then translates it ten units right. After 150 moves the point is (p, q). Find $\lfloor |p| + |q| \rfloor$.

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2008 AIME II Problem 10: Increasing distances on a small grid

On the sixteen points $\{0, 1, 2, 3\}^2$, a path visits distinct points with strictly increasing successive distances. Let m be its maximum number of points and r the number of paths attaining that maximum. Find mr.

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2008 AIME II Problem 11: Two tangent circles along an isosceles base

Triangle ABC has $AB = AC = 100$, $BC = 56$. A radius-16 circle touches AC and BC. Another circle lies entirely inside the triangle, touches AB and BC, and is externally tangent to the first circle. Its radius is $m - n\sqrt{k}$ with squarefree k. Find $m + nk$.

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2008 AIME II Problem 13: Inverting the exterior of a regular hexagon

A regular hexagon centered at 0 in the complex plane has opposite sides one unit apart, including a pair parallel to the imaginary axis. Let $S = \{1/z : z \text{ is outside the hexagon}\}$. If its area is $a\pi + \sqrt{b}$, find $a + b$.

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2008 AIME II Problem 14: An equilateral constraint inside a rectangle

For $a \geq b > 0$, suppose $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ has a solution with $0 \leq x < a$, $0 \leq y < b$. Let ρ be the largest possible a/b. If $\rho^2 = m/n$ in lowest terms, find $m + n$.

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2009 AIME I Problem 4: A diagonal meeting a short intercept segment

In parallelogram $ABCD$, points $M \in AB$ and $N \in AD$ satisfy $AM/AB = 17/1000$ and $AN/AD = 17/2009$. If $P = AC \cap MN$, find AC/AP.

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2009 AIME I Problem 5: A midpoint reflection and an angle bisector

In triangle ABC, $AC = 450$, $BC = 300$. Let K be the midpoint of AC, and let the angle bisector CL meet AB at L. Put $P = BK \cap CL$ and reflect P across K to obtain M. If $AM = 180$, find LP.

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2009 AIME I Problem 11: Integral triangle areas on a line

Choose distinct nonnegative lattice points P, Q on $41x + y = 2009$. Count triangles OPQ of positive integer area, where $O = (0, 0)$.

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2009 AIME I Problem 12: A universal perimeter ratio from an altitude circle

Right triangle ABC has $AC = 12$, $BC = 35$, and altitude CD to hypotenuse AB. The circle with diameter CD has nonbase tangents from A and B meeting at I outside ABC. Write the perimeter of ABI divided by AB as reduced m/n, and find $m + n$.

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2009 AIME I Problem 15: A fixed-angle arc from two incenter circles

Triangle ABC has $AB = 10$, $BC = 14$, $CA = 16$. For D inside BC, let I_B, I_C be the incenters of ABD, ACD. The circles through B, I_B, D and through C, I_C, D meet again at P. The maximum area $[BPC]$ is $a - b\sqrt{c}$ in reduced squarefree form. Find $a + b + c$.

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2009 AIME II Problem 3: Perpendicular lines in a rectangle

Rectangle $ABCD$ has $AB = 100$. If E is the midpoint of AD and $AC \perp BE$, find the greatest integer less than AD.

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2009 AIME II Problem 5: A circle tangent to three interior circles

An equilateral triangle is inscribed in a radius-10 circle. Three internally tangent circles touch the outer circle at the triangle vertices and have radii $3, 2, 2$. A fourth circle is externally tangent to all three. Its radius is reduced m/n. Find $m + n$.

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2009 AIME II Problem 10: Two equal-angle observations

Points A, B, C, D satisfy $AB = 5$, $BC = 12$, $AC = 13$, $\angle BAD = \angle CAD$, and $\angle ACB = \angle DCB$. Write $AD = p\sqrt{q}/r$ in reduced squarefree form and find $p + q + r$.

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2009 AIME II Problem 13: A product of semicircle chords

A radius-2 semicircle with diameter endpoints A, B is divided into seven equal arcs by $C_1, \ldots, C_6$. Let $N = \prod_{k=1}^{6} AC_k \, BC_k$. Find N modulo 1000.

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2009 AIME II Problem 15: Maximum separation of two chord intercepts

Circle diameter MN has length one. On one semicircle, A is the arc midpoint and $MB = 3/5$. A variable point C lies on the other semicircle. The chords AC, BC meet MN at D, E. If the largest DE is $r - s\sqrt{t}$ in reduced squarefree form, find $r + s + t$.

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2010 AIME I Problem 8: The smallest disk enclosing floor-defined squares

Let $R = \{(x, y) : \lfloor x \rfloor^2 + \lfloor y \rfloor^2 = 25\}$. The smallest closed disk containing R has radius $\sqrt{m}/n$ in reduced squarefree form. Find $m + n$.

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2010 AIME I Problem 11: Revolving a triangular inequality region

The region satisfying $|8 - x| + y \le 10$ and $3y - x \ge 15$ is revolved around $3y - x = 15$. Write its volume as $m\pi/(n\sqrt{p})$ in reduced squarefree form and find $m + n + p$.

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2010 AIME I Problem 13: A rectangle and semicircle cut in a fixed area ratio

A rectangle $ABCD$ and a semicircle with diameter AB lie on opposite sides of AB. A line meets the semicircle at N, AB at U, and CD at T, dividing their union in area ratio $1 : 2$. Given $AU = 84$, $UB = 168$, $AN = 126$, write $AD = m\sqrt{n}$ in squarefree form and find $m + n$.

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2010 AIME I Problem 15: Equal inradii determine a cevian length

Triangle ABC has $AB = 12$, $BC = 13$, $AC = 15$. A point M inside AC makes triangles ABM and BCM have equal inradii. Write $AM/CM = p/q$ in lowest terms and find $p + q$.

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2010 AIME II Problem 2: Distance to the nearest side

Choose a point uniformly inside a unit square. Its distance to the nearest side is d. Find numerator plus denominator of $P(1/5 \le d \le 1/3)$.

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2010 AIME II Problem 9: A regular hexagon bounded by rotated chords

In regular hexagon $ABCDEF$, let G, H, I, J, K, L be the midpoints of AB, BC, CD, DE, EF, FA. The segments AH, BI, CJ, DK, EL, FG bound a smaller regular hexagon. If its area divided by the original area is reduced m/n, find $m + n$.

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2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter

Two noncongruent integer-sided isosceles triangles have equal areas and perimeters, and their base lengths are in ratio $8:7$. Find their least possible common perimeter.

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2010 AIME II Problem 14: A doubled angle in a right triangle

Right triangle ABC has right angle C, $AB = 4$, and $\angle A < 45^\circ$. A point P on AB satisfies $CP = 1$ and $\angle APC = 2\angle ACP$. If $AP/BP = p + q\sqrt{r}$ in squarefree form, find $p + q + r$.

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2010 AIME II Problem 15: A radical axis evaluated by endpoint powers

Triangle ABC has $AC = 13$, $BC = 14$, $AB = 15$. Let M,N be the midpoints of AC,AB, and let the angle bisectors from B,C meet AC,AB at D,E. The circles AMN and ADE meet again at P. Ray AP meets BC at Q. If $BQ/CQ = m/n$ in lowest terms, find $m - n$.

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2011 AIME I Problem 2: Parallel segments across a rectangle

In rectangle $ABCD$, $AB = 12$ and $BC = 10$. Interior points E, F satisfy $BE = 9$, $DF = 8$, $BE \parallel DF$, and $EF \parallel AB$; the line BE meets side AD. Write $EF = m\sqrt{n} - p$ with squarefree n and positive integers m, n, p, and find $m + n + p$.

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2011 AIME I Problem 3: Coordinates in rotated perpendicular axes

Line L has slope $5/12$ and contains $(24, -1)$. Its perpendicular M contains $(5, 6)$. Use L, M as the new coordinate axes, with positive directions toward these respective points. If $(-14, 27)$ has coordinates (α, β) in this system, find $\alpha + \beta$.

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2011 AIME I Problem 4: Feet on angle bisectors

Triangle ABC has $AB = 125$, $AC = 117$, and $BC = 120$. Let M, N be the perpendicular projections of C onto the internal angle bisectors at B, A, respectively. Find MN.

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2011 AIME I Problem 8: Folding triangular legs of equal height

A triangular sheet has side lengths $23, 27, 30$. Cut off a corner triangle at each vertex by a fold line parallel to the opposite side, with the three corner interiors disjoint. Fold these corners downward through 90° so their vertex tips lie on one plane below the central tabletop. What is the greatest height, written $m\sqrt{n}/p$ in reduced form with squarefree n? Find $m + n + p$.

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2011 AIME I Problem 10: Obtuse triangles in a regular polygon

Three vertices are chosen uniformly from a regular n-gon. Their triangle is obtuse with probability $93/125$. Find the sum of all possible n.

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2011 AIME I Problem 13: The height of a tilted cube

A cube of edge length 10 lies above a plane. Its nearest vertex is A, and the three vertices adjacent to A are at distances $10, 11, 12$ from the plane. Express the distance of A as $(r - \sqrt{s})/t$ in reduced form and find $r + s + t$.

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2011 AIME I Problem 14: Projecting the midpoint square of an octagon

In a regular octagon $A_1 \cdots A_8$, let M_i be the midpoint of $A_i A_{i+1}$ for odd i. Inward rays from M_1, M_3, M_5, M_7 are perpendicular to their neighboring rays. Their successive intersections are B_1, B_3, B_5, B_7, with $B_1 \in R_1 \cap R_3$. If $B_1 B_3$ equals an octagon side, write $\cos(2\angle A_3 M_3 B_1) = m - \sqrt{n}$ and find $m + n$.

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2011 AIME II Problem 2: Three equal segments spanning a square

In square $ABCD$, points E and F lie in the interiors of AD and BC, respectively. If $BE = EF = FD = 30$, find the area of the square.

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2011 AIME II Problem 3: Integer angles in arithmetic progression

The interior angle measures of a convex 18-gon, in increasing order, form a strictly increasing arithmetic progression of integers. Find its smallest angle in degrees.

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2011 AIME II Problem 4: A midpoint on an angle bisector

In triangle ABC, $AB = 20$ and $AC = 11$. The internal bisector from A meets BC at D, and M is the midpoint of AD. Line BM meets AC at P. If $CP/PA = m/n$ in lowest terms, find $m + n$.

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2011 AIME II Problem 8: Independent choices among rotated complex roots

Let $z_1, \ldots, z_{12}$ be the roots of $z^{12} = 2^{36}$. Independently choose each w_j to equal z_j or iz_j. If the maximum of $\operatorname{Re} \sum_j w_j$ is $m + \sqrt{n}$ for positive integers m, n, find $m + n$.

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2011 AIME II Problem 10: A diameter through two chord projections

In a circle of radius 25 centered at O, chords of lengths 30 and 14 meet at P. Their midpoints are 12 units apart. If $OP^2 = m/n$ in lowest terms, find the remainder of $m + n$ modulo 1000.

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2011 AIME II Problem 13: Circumcenters along a square diagonal

Point P lies inside diagonal AC of square $ABCD$, with $AP > CP$ and side length 12. Let O_1, O_2 be the circumcenters of ABP, CDP. If $\angle O_1PO_2 = 120^\circ$ and $AP = \sqrt{a} + \sqrt{b}$ for positive integers a, b, find $a + b$.

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2012 AIME I Problem 8: A cube slice is a truncated tetrahedron

Take the unit cube $[0, 1]^3$ and cut it by the plane through $(0, 0, 0)$, $(1/2, 1, 0)$, and $(1, 0, 1/2)$. These coordinates specify the vertex and edge-midpoints of the archived diagram. If the larger part has reduced volume p/q, find $p + q$.

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2012 AIME I Problem 11: Reachability in a lattice semigroup

From $(0, 0)$ allow jumps $(7, 2), (2, 7), (-5, -10), (-10, -5)$. Let M count reachable lattice points with $|x| + |y| \le 100$. Find M mod 1000.

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2012 AIME I Problem 12: An angle bisector inside a trisected right angle

Triangle ABC is right at C. Rays CD, CE trisect that right angle and meet AB in the order A, D, E, B. Given $DE/EB = 8/15$, write $\tan B = m\sqrt{p}/n$ in reduced squarefree form and find $m + n + p$.

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2012 AIME I Problem 13: Equilateral triangles with three prescribed center distances

Three circles have a common center and radii $3, 4, 5$. Among equilateral triangles having one vertex on each circle, maximize the area. Write it as $a + (b/c)\sqrt{d}$, with positive integers, b, c coprime and d squarefree, and find $a + b + c + d$.

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2012 AIME I Problem 14: A zero root sum places the triangle centroid at the origin

The complex roots a, b, c of $z^3 + qz + r$ form a nondegenerate right triangle in the complex plane and satisfy $|a|^2 + |b|^2 + |c|^2 = 250$. If its hypotenuse is h, find h^2. The coefficients q, r need not be real.

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2012 AIME II Problem 4: Three equal travel times determine a rectangle

From a rectangle's northeast corner, riders of speeds 8.6, 6.2, and 5 m/s start together. The first follows the boundary westward, the second southward, and the third rides straight to a point D on the south side. They first reach D simultaneously. The east-west side is longer. Find the primitive ratio of length, width, and distance from the southeast corner to D, and report its three entries' sum.

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2012 AIME II Problem 5: A four-flap net folds into a square pyramid

Concentric, parallel-sided squares have side lengths 40 and 15. Attach to each side of the smaller square the triangle whose apex is the midpoint of the corresponding outer side. Fold the four triangles into the lateral faces of a pyramid on the smaller square. Find its volume.

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2012 AIME II Problem 6: A complex-distance maximum fixes a fourth power

Choose $z = a + bi$ with $|z| = 5$ and $b > 0$ to maximize $|(1 + 2i)z^3 - z^5|$. If $z^4 = c + di$, find $c + d$.

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2012 AIME II Problem 13: Sum squared distances before computing rotated vertices

An equilateral triangle ABC has side $\sqrt{111}$. The two points D_1, D_2 satisfy $AD_i = \sqrt{111}$ and $BD_i = \sqrt{11}$. For each segment AD_i, construct both equilateral third vertices, giving four points $E_1, \ldots, E_4$. Find $\sum_{j=1}^{4} CE_j^2$.

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2012 AIME II Problem 15: Two circle equations and their common chord

A triangle with $AB = 5$, $AC = 3$, $BC = 7$ has circumcircle ω. Its internal angle bisector from A meets BC at D and ω again at E. The circle with diameter DE meets ω again at $F \neq E$. Write $AF^2 = m/n$ in lowest terms and find $m + n$.

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2013 AIME I Problem 3: Area fractions determine a reciprocal side ratio

Two perpendicular cuts parallel to the sides partition a square into two squares and two nonsquare rectangles. Let a, b be the two segment lengths into which either cut divides its perpendicular side. The two smaller squares occupy $9/10$ of the original area. Find $a/b + b/a$.

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2013 AIME I Problem 7: Triangle through three face centers

A rectangular box has side lengths $12, 16, h$, with $h > 0$. The centers of the three faces incident to one vertex form a triangle of area 30. Write $h = m/n$ in lowest terms and find $m + n$.

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2013 AIME I Problem 9: A folding crease from equal distances

An equilateral triangle ABC has side 12. Reflect A onto the point D of BC with $BD = 9$. Find the length of the crease inside the triangle as $m\sqrt{p}/n$, with m, n coprime and p squarefree, and report $m + n + p$.

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2013 AIME I Problem 12: A supporting line fixes a hexagon enclosure

A regular hexagon $ABCDEF$ of side one is contained in a triangle PQR with angles $P = 75^\circ$, $Q = 60^\circ$, $R = 45^\circ$. Its edges AB, CD lie on PQ, QR, and another vertex lies on RP. Write the triangle area as $(a + b\sqrt{c})/d$, with c squarefree and a, d coprime, and find $a + b + c + d$.

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2013 AIME I Problem 13: A self-similar union sums to one rational area

Begin with a triangle AB_0C_0 of side lengths $AB_0 = 12$, $B_0C_0 = 17$, $AC_0 = 25$. At each stage choose B_n on AB_{n-1} and C_n on AC_{n-1} so that, in the displayed vertex order,

$$AB_nC_n \sim B_{n-1}C_nC_{n-1} \sim AB_{n-1}C_{n-1}.$$

If the union of triangles $B_{n-1}C_nB_n$ for $n \geq 1$ has reduced rational area p/q, determine q.

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2013 AIME II Problem 4: Choose the correct equilateral vertex by rotation

Let $A = (1, 0)$ and $B = (2, 2\sqrt{3})$. Of the two equilateral triangles on AB, choose the one with third vertex C in the first quadrant. If its center is (x, y) and $xy = p\sqrt{q}/r$ in reduced squarefree form, find $p + q + r$.

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2013 AIME II Problem 5: An angle sine from one small triangle area

In an equilateral triangle ABC, the points D, E divide BC into three equal parts. Express $\sin\angle DAE$ as $a\sqrt{b}/c$, with a, c coprime and b squarefree, and find $a + b + c$.

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2013 AIME II Problem 8: Repeated chord lengths force a diameter

A convex hexagon inscribed in a circle has successive side lengths $22, 22, 20, 22, 22, 20$. Its circumradius is $p + \sqrt{q}$, for positive integers p, q. Find $p + q$.

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2013 AIME II Problem 10: A constant area ratio turns a secant maximum into a right angle

A circle has center O and radius $\sqrt{13}$. A point A has $AO = 4 + \sqrt{13}$, and B is the point of the circle closest to A. As a secant through A cuts the circle at K, L, maximize $[BKL]$. Write the maximum as $(a - b\sqrt{c})/d$ in reduced squarefree form and find $a + b + c + d$.

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2013 AIME II Problem 13: Affine midpoint coordinates linearize two distances

In an isosceles triangle ABC with $AC = BC$, choose D on BC so that $CD = 3BD$, and let E be the midpoint of AD. Given $CE = \sqrt{7}$ and $BE = 3$, express $[ABC] = m\sqrt{n}$ with n squarefree and find $m + n$.

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2013 AIME II Problem 15: A triangle identity with a radical-sign branch condition

Triangle angles A, B, C satisfy

$$\cos^2 A + \cos^2 B + 2\sin A \sin B \cos C = 15/8,$$

$$\cos^2 B + \cos^2 C + 2\sin B \sin C \cos A = 14/9.$$

The expression $\cos^2 C + \cos^2 A + 2\sin C \sin A \cos B$ has the form $(p - q\sqrt{r})/s$ for positive integers p, q, r, s, with r squarefree and $\gcd(p+q, s) = 1$. Determine $p + q + r + s$.

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2014 AIME I Problem 1: The length of a crossed shoelace

Eight eyelets lie at four equally spaced positions on each long side of a 50 by 80 mm rectangle. A lace joins the two eyelets at one short end, then has two crossing strands between each successive eyelet pair. Each free end needs another 200 mm. Find its minimum length.

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2014 AIME I Problem 5: Disk-separable vertex subsets

Count subsets of a regular dodecagon that are precisely its vertices strictly inside some circle.

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2014 AIME I Problem 7: The largest argument seen from an exterior point

Complex numbers satisfy $|w| = 1$ and $|z| = 10$. Let $\theta = \arg((w - z)/z)$. If the maximum of $\tan^2 \theta$ is p/q in lowest terms, find $p + q$.

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2014 AIME I Problem 10: A disk rolling around a larger disk

A unit disk rolls without slipping around the outside of a fixed radius-5 disk. Initially their tangency point is A and the larger center is E. After the unit disk makes one full rotation relative to the plane, its marked point initially at A is B. Write $\sin^2 \angle BEA = m/n$ in lowest terms and find $m + n$.

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2014 AIME I Problem 11: Diagonal endpoint of a random walk

A planar simple random walk starts at the origin and takes six unit axis steps. Find numerator plus denominator of its probability of ending on either diagonal $y = x$ or $y = -x$.

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2014 AIME I Problem 13: A square cut by equal perpendicular chords

Points E, F, G, H lie respectively on sides AB, BC, CD, DA of a square. Segments EG, FH are perpendicular, each of length 34, and meet at P. The areas $AEPH, BFPE, CGPF, DHPG$ have ratio $269 : 275 : 405 : 411$. Find the square area.

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2014 AIME I Problem 15: A circle through a right-triangle vertex

In right triangle ABC, $AB = 3, BC = 4, AC = 5$. A circle meets AB at B, E, BC at B, D, and AC at F, G. Suppose $EF = DF$ and $DG/EG = 3/4$. Write $DE = a\sqrt{b}/c$ in reduced radical form and find $a + b + c$.

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2014 AIME II Problem 3: A hinged rectangle retaining its area

A rectangle has sides a and 36. Hinges at the vertices and midpoints of the 36-sides allow it to become a convex hexagon: the length-a sides remain parallel 24 units apart, and each former 36-side becomes two 18-unit segments meeting outward. If area is unchanged, find a^2.

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2014 AIME II Problem 8: Three mutually tangent circles and a diameter

A radius-2 circle has diameter AB. A smaller circle is internally tangent at A. A third circle is internally tangent to the large circle, externally tangent to the second circle, and tangent to AB. The second radius is three times the third. Write it as $\sqrt{m} - n$ and find $m + n$.

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2014 AIME II Problem 10: A reciprocal equation creates an equilateral triangle

Let $|z| = 2014$. Form the polygon with vertices z and all complex w satisfying $1/(z + w) = 1/z + 1/w$. Its area is $n\sqrt{3}$. Find n modulo 1000.

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2014 AIME II Problem 11: An isosceles extension of a triangle

In triangle RED, angles at R, E are 75 and 45 degrees and $RD = 1$. Let M be the midpoint of RD, and choose C on ED with $RC \perp EM$. Extend DE beyond E to A so that $AC = AR$. If $AE = (a - \sqrt{b})/c$ in reduced form, find $a + b + c$.

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2014 AIME II Problem 12: A triple-angle condition forces one angle

Triangle angles satisfy $\cos 3A + \cos 3B + \cos 3C = 1$. Two sides have lengths 10 and 13. The greatest possible third side is $\sqrt{m}$; find m.

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2014 AIME II Problem 14: Midpoint projections on an angle bisector

Triangle ABC has $AB = 10$, angle $A = 30°$, and angle $C = 45°$. On line BC, let H be the foot from A, D the internal angle-bisector intersection, and M the midpoint of BC. Let N be the midpoint of HM, and choose P on ray AD with $PN \perp BC$. If $AP^2 = m/n$ in lowest terms, find $m + n$.

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2015 AIME I Problem 4: Midpoints between adjacent equilateral triangles

Collinear points A, B, C occur in that order, with $AB = 16$ and $BC = 4$. Equilateral triangles ABD and BCE lie on the same side of the line. Let M, N be the midpoints of AE, CD. If $[BMN] = x$, find x^2.

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2015 AIME I Problem 6: Two equal-step arcs sharing endpoints

On one circle, A, B, C, D, E occur at equal steps along a minor arc. On a circle centered at C, points E, F, G, H, I, A occur at equal steps along a minor arc. If $\angle ABD - \angle AHG = 12^\circ$, find $\angle BAG$ in degrees.

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2015 AIME I Problem 7: Two nested squares in a square

In square $ABCD$, let E be the midpoint of AD. A square $FGHJ$ has F, G on CE, H on AB, and J on BC. A square $KLMN$ has K, L on GH, M on AD, and N on AB. If its area is 99, find the area of $FGHJ$.

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2015 AIME I Problem 11: Integral isosceles triangles with a fixed incenter distance

An isosceles triangle has $AB = AC$, positive integer side lengths, and incenter I satisfying $BI = 8$. Find its minimum perimeter.

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2015 AIME I Problem 15: The area of an oblique cylinder section

A solid cylinder has radius 6 and height 8. A plane passes through its center and two points on one rim whose smaller arc measures 120 degrees. The plane section has area $a\pi + b\sqrt{c}$, with positive integers a, b and squarefree c. Find $a + b + c$.

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2015 AIME II Problem 4: Logarithmic lengths form a familiar triangle

An isosceles trapezoid has bases $\log 3, \log 192$ and altitude $\log 16$, with a common logarithm base greater than 1. Its perimeter is $\log(2^p 3^q)$. Find $p + q$.

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2015 AIME II Problem 5: Adjacent cells in a random pair

Two different cells are chosen uniformly from an n by n square grid. Find the least n for which their probability of sharing an edge is below $1/2015$.

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2015 AIME II Problem 7: The area polynomial of an inscribed rectangle

Triangle ABC has $AB = 12$, $BC = 25$, $CA = 17$. A rectangle has two adjacent vertices P, Q on AB, AC and its opposite side on BC. If $PQ = \omega$ and its area is $\alpha\omega - \beta\omega^2$, write $\beta = m/n$ in lowest terms and find $m + n$.

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2015 AIME II Problem 9: A cube corner enters a circular opening

A water-filled cylinder has radius 4 and height 10. A cube of side 8 is lowered with a body diagonal on the cylinder axis until it first meets the rim. If the displaced water volume is v, find v^2.

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2015 AIME II Problem 11: A line through the circumcenter perpendicular to a radius

An acute triangle has $AB = 5$, $BC = 4$, and circumcenter O. The line through O perpendicular to OB meets lines AB, BC at P, Q, with $BQ = 9/2$. If $BP = m/n$ in lowest terms, find $m + n$.

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2015 AIME II Problem 15: Equal areas cut from two tangent circles

Two externally tangent circles have radii 1 and 4 and meet at A. A common external tangent touches them at B, C. A line through A meets the circles again at D, E, respectively. If B, C lie on the same side of the line and $[DBA] = [ACE] = m/n$ in lowest terms, find $m + n$.

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2016 AIME I Problem 3: Downward simple paths on an icosahedron

Orient an icosahedron with top and bottom vertices and two horizontal pentagons. Count vertex-simple edge paths from top to bottom that never move upward.

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2016 AIME I Problem 4: A dihedral angle in a hexagonal prism

A right prism has regular hexagonal bases of side 12 and height h. A tetrahedron consists of a vertex A and its three adjacent prism vertices. Its face in the base and its face opposite A meet at a 60-degree dihedral angle. Find h^2.

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2016 AIME I Problem 6: The arc midpoint and the incenter

The internal angle bisector from C in triangle ABC meets AB at L and the circumcircle again at D. The incenter is I, with $LI = 2$ and $LD = 3$. Write $CI = p/q$ in lowest terms and find $p + q$.

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2016 AIME I Problem 9: A rectangle optimization with a segment inconsistency

Triangle ABC has $AB = 40$, $AC = 31$, and $\sin A = 1/5$. It lies in rectangle $AQRS$ with B on segment QR and C on segment RS. Determine the maximum rectangle area. The reported contest answer is 744; the segment conditions require the correction below.

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2016 AIME I Problem 14: Lattice shapes intersected by a rational line

At every lattice point place a radius $1/10$ circle and an axis-parallel side $1/5$ square. Count the total shapes meeting the segment from $(0, 0)$ to $(1001, 429)$.

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2016 AIME I Problem 15: A tangent configuration gives a difference of squares

Circles meet at X, Y and share an external tangent touching them at A, B, closer to X than Y. A circle through A, B meets the first again at D and the second again at C, where D, Y, C are collinear. Given $XD = 37$, $XY = 47$, $XC = 67$, find AB^2.

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2016 AIME II Problem 5: An infinite chain of right-triangle altitudes

Right triangle ABC_0 has pairwise coprime positive integer sides and perimeter p. Let C_1 be the foot from C_0 to AB; for $n \geq 2$, let C_n be the foot from C_{n-1} to $C_{n-2}B$. If $C_0C_1 + C_1C_2 + \cdots = 6p$, find p.

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2016 AIME II Problem 7: Three nested squares with two integer areas

Two concentric squares have parallel sides. The outer area is 2016 and the inner area is a smaller positive integer. A third square has one vertex on each outer side, while each inner vertex lies on one of its sides. Find the difference between the greatest and least possible positive integer areas of this third square.

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2016 AIME II Problem 10: A chord length from a ratio of subtriangle areas

In triangle ABC on a circle, points P, Q lie on AB in the order A, P, Q, B, with $AP = 4$, $PQ = 3$, $QB = 6$. Rays CP, CQ meet the circle again at S, T. If $AS = 7$ and $BT = 5$, write $ST = m/n$ in lowest terms and find $m + n$.

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2016 AIME II Problem 14: A common sphere through a triangular bipyramid

An equilateral triangle ABC has side 600. Points P, Q on opposite sides of its plane are each equidistant from A, B, C. The face half-planes PAB, QAB form a 120-degree dihedral angle. A point O is distance d from all five vertices. Find d.

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2017 AIME I Problem 1: Triangles from boundary points

A triangle has 3 additional points on one side, 4 on another, and 5 on the third. Count nondegenerate triangles whose vertices are among all 15 marked points.

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2017 AIME I Problem 4: Equal lateral edges above an isosceles base

A tetrahedron has base side lengths 20,20,24 and all three remaining edges of length 25. Its volume is $m\sqrt{n}$ with squarefree positive integer n. Find $m+n$.

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2017 AIME I Problem 6: A chord intersecting an isosceles triangle

An inscribed isosceles triangle has equal angles x degrees. A chord with two independent uniform endpoints intersects the triangle with probability $14/25$. Find the difference between possible x values.

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2017 AIME I Problem 8: Two random right-triangle vertices

On the same semicircle of diameter $OP = 200$, select Q, R so that angles POQ and POR are independently uniform between 0 and 75 degrees. Find numerator plus denominator of $P(QR \le 100)$.

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2017 AIME I Problem 10: A real cross ratio traces a circle

Let $z_1 = 18 + 83i$, $z_2 = 18 + 39i$, and $z_3 = 78 + 99i$. Among complex $z \neq z_3$ making $\frac{z_3 - z_1}{z_2 - z_1}\frac{z - z_2}{z - z_3}$ real, take the one with greatest imaginary part. Find its real part.

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2017 AIME I Problem 15: The smallest equilateral triangle between three sides

A right triangle has legs 5 and $2\sqrt{3}$. An equilateral triangle has one vertex on each side. Its least possible area is $m\sqrt{p}/n$, where m/n is reduced and p squarefree. Find $m+n+p$.

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2017 AIME II Problem 3: Voronoi region inside a triangle

A point is uniform in the triangle with vertices $(0,0), (12,0), (8,10)$. Find numerator plus denominator of its probability of being closest to $(12,0)$.

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2017 AIME II Problem 10: An area bisector in a cut rectangle

Rectangle $ABCD$ has $AB = 84$, $AD = 42$. Let M be the midpoint of AD, N the point on AB with $AN = 28$, and $O = CM \cap DN$. A point P on the boundary of $BCON$ makes segment BP bisect its area. Find $[CDP]$.

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2017 AIME II Problem 12: The limit point of rotating nested circles

Starting with a unit circle centered at O, choose A_0 on it. For $n \geq 1$, a circle of radius r^n is internally tangent to the preceding circle at A_{n-1}, and A_n is 90 degrees counterclockwise from that tangency point on the new circle. For $r = 11/60$, write the distance from O to the common limit point as m/n and find $m + n$.

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2017 AIME II Problem 13: Isosceles triangles in regular polygons

Let $f(n)$ count isosceles vertex triangles of a regular n-gon, including equilateral ones. Find the sum of $n \geq 3$ satisfying $f(n+1) - f(n) = 78$.

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2017 AIME II Problem 14: Lines containing exactly eight lattice points

In $\{1, \ldots, 10\}^3$ count lines containing exactly 8 grid points.

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2017 AIME II Problem 15: The geometric median of a disphenoid

A tetrahedron has opposite edge pairs $AD = BC = 28$, $AC = BD = 44$, and $AB = CD = 52$. If the least possible value of $AX + BX + CX + DX$ over points X in space is $m\sqrt{n}$ with squarefree n, find $m + n$.

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2018 AIME I Problem 4: An equal-sided chain inside an isosceles triangle

Triangle ABC has $AB = AC = 10$ and $BC = 12$. Interior side points $D \in AB$, $E \in AC$ satisfy $AD = DE = EC$. Write $AD = p/q$ in lowest terms and find $p + q$.

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2018 AIME I Problem 7: Isosceles triangles in a hexagonal prism

A right regular hexagonal prism has base side 1 and height 2. Count isosceles triangles among its 12 vertices.

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2018 AIME I Problem 8: The largest disk in an equiangular hexagon

A convex equiangular hexagon $ABCDEF$ has successive side lengths $AB = 6$, $BC = 8$, $CD = 10$, $DE = 12$. If the largest disk inside it has diameter d, find d^2.

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2018 AIME I Problem 13: The area between two subtriangle incenters

Triangle ABC has $AB = 30$, $BC = 32$, $AC = 34$. For X interior to BC, let I_1, I_2 be the incenters of ABX, ACX. Find the least possible area of AI_1I_2.

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2018 AIME I Problem 15: Three cyclic side orders share an area invariant

Four distinct stick lengths form three noncongruent convex cyclic quadrilaterals, all of circumradius 1 and common area K. The sines of their acute diagonal angles are $2/3, 3/5, 6/7$. Write $K = m/n$ in lowest terms and find $m + n$.

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2018 AIME II Problem 4: Six regions of a crossed octagon

An equiangular octagon has cyclic labels C, A, R, O, L, I, N, E, with alternating side lengths $\sqrt{2}, 1$, starting at CA. Joining the same vertices in the order C, O, R, N, E, L, I, A forms six disjoint triangular regions. If their total area is a/b in lowest terms, find $a + b$.

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2018 AIME II Problem 7: Rational lengths among equal-area parallel slices

A triangle has sides $AB = 9$, $AC = 12$, $BC = 5\sqrt{3}$. Lines parallel to BC divide it into 2450 equal-area regions. Let the kth parallel segment, including BC as the 2450 th, bound the first k regions nearest A. How many of these 2450 segments have rational length?

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2018 AIME II Problem 9: A fan of triangle centroids is a scaled midpoint polygon

An octagon has vertices $A = (0, 6), B = (8, 0), C = (19, 0), D = (27, 6), E = (27, 17), F = (19, 23)$, $G = (8, 23), H = (0, 17)$. Let J be the midpoint of AH. The segments from J to B through G form seven triangles. Find the area of the convex polygon formed by their centroids.

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2018 AIME II Problem 12: An area balance bisects one diagonal

A convex quadrilateral has $AB = CD = 10$, $BC = 14$, $AD = 2\sqrt{65}$. Its diagonals meet at P, with $[APB] + [CPD] = [BPC] + [APD]$. Find its area.

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2018 AIME II Problem 14: A tangent through a near chord endpoint

The incircle of triangle ABC touches BC at X. Line AX meets it again at Y. A tangent at Y meets segments AB,AC at P,Q. Given $AP = 3$, $PB = 4$, $AC = 8$, write $AQ = m/n$ in lowest terms and find $m + n$.

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2019 AIME I Problem 2: A gap between ordered samples

Choose distinct J, B uniformly from [20] in order. Find numerator plus denominator of $P(B - J \geq 2)$.

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2019 AIME I Problem 3: An inset hexagon in a right triangle

In right triangle PQR, $PR = 15$, $QR = 20$, $PQ = 25$. Points A,B lie on PQ, C,D on QR, and E,F on PR, in cyclic order around the triangle, with $PA = QB = QC = RD = RE = PF = 5$. Find the area of hexagon ABCDEF.

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2019 AIME I Problem 6: Two perpendiculars turn a product into a square

In convex quadrilateral KLMN, $MN \perp KM$ and $KL \perp LN$. The perpendicular from L to KN meets KM at O. Given $MN = 65$, $KL = 28$, and $KO = 8$, find MO.

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2019 AIME I Problem 11: A circle tangent to all three excircles

An integer-sided triangle has $AB = AC$. A circle centered at its incenter is internally tangent to the A-excircle and externally tangent to the other two excircles. Find the smallest possible perimeter.

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2019 AIME I Problem 12: A right angle becomes a purely imaginary quotient

Let $f(z) = z^2 - 19z$. Complex numbers z,f(z),f(f(z)) form a nondegenerate right triangle, with right angle at f(z). One admissible z is $m + \sqrt{n} + 11i$ for positive integers m,n, with n squarefree. Find $m + n$.

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2019 AIME I Problem 13: Two cyclic quadrilaterals share a radical axis

Triangle ABC has $AB = 4$, $BC = 5$, $CA = 6$. Points A,B,D,E occur in this order on a line. The circumcircles of ACD and EBC meet again at F, with $DF = 2$, $EF = 7$. Write $BE = (a + b\sqrt{c})/d$ in reduced form with c squarefree; find $a + b + c + d$.

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2019 AIME I Problem 15: Two internally tangent circles locate a chord midpoint

P lies on chord AB of a circle, with $AP = 5$, $PB = 3$. A circle through A,P is internally tangent to the original circle at A; another through B,P is internally tangent at B. Their other intersection is Q. The line PQ cuts an 11-unit chord XY of the original circle. If $PQ^2 = m/n$ reduced, find $m + n$.

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2019 AIME II Problem 1: The overlap of two congruent triangles

Congruent triangles ABC and BAD lie on the same side of AB, with C distinct from D, $AB = 9$, $BC = AD = 10$, $CA = DB = 17$. If their intersection area is m/n in lowest terms, find $m + n$.

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2019 AIME II Problem 7: Three parallel cuts determine a homothetic triangle

Triangle ABC has $AB = 120$, $BC = 220$, $CA = 180$. Three lines cut interior segments parallel to BC,CA,AB of lengths 55,45,15 respectively. Find the perimeter of the triangle formed by the three lines.

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2019 AIME II Problem 11: Two tangent circles and a median

Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. One circle passes through B and is tangent to AC at A; another passes through C and is tangent to AB at A. Their other intersection is K. If $AK = m/n$ reduced, find $m + n$.

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2019 AIME II Problem 13: Curved sector areas are affine in the interior point

A regular octagon $A_1 \dots A_8$ is inscribed in a circle of area 1. An interior point P forms curved regions bounded by PA_i, PA_{i+1} and the minor arc between consecutive vertices. Regions 12 and 34 have areas 1/7 and 1/9. Region 67 has area $1/8 - \sqrt{2}/n$. Find n.

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2019 AIME II Problem 15: Two chord powers recover both sides of a triangle

In acute triangle ABC, P is the perpendicular foot from C to AB and Q the foot from B to AC. Line PQ meets the circumcircle at X,Y in the order X,P,Q,Y, with $XP = 10$, $PQ = 25$, $QY = 15$. If $AB \cdot AC = m\sqrt{n}$ with n squarefree, find $m + n$.

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2020 AIME I Problem 1: Three equal-length steps force a seventh of a straight angle

In isosceles triangle ABC with $AB = AC$, interior side points $D \in AC$ and $E \in AB$ satisfy $AE = ED = DB = BC$. Write angle ABC as m/n degrees in lowest terms and find $m + n$.

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2020 AIME I Problem 6: Equal spheres resting in unequal circular holes

Two circular holes of radii 1 and 2 have centers 7 units apart in a horizontal board. Two equal spheres rest in the holes, with their centers above the board, and touch each other. If their common radius squared is m/n reduced, find $m + n$.

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2020 AIME I Problem 8: A shrinking spiral is a complex geometric series

A bug starts at O and walks 5 units east. Before each later walk it turns 60 degrees counterclockwise and halves its walking distance. Its positions converge to P. If $OP^2 = m/n$ reduced, find $m + n$.

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2020 AIME I Problem 13: A perpendicular bisector cuts two angle bisectors

Triangle ABC has $AB = 4$, $BC = 5$, $CA = 6$. Its angle bisector AD meets BC at D. The perpendicular bisector of AD meets the internal angle-bisector lines from B,C at E,F. If $[AEF] = m\sqrt{n}/p$ reduced with n squarefree, find $m + n + p$.

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2020 AIME I Problem 15: An orthocenter tangent determines a chord and an altitude

An acute triangle ABC has orthocenter H and circumcircle omega. The tangent at H to the circumcircle of HBC meets omega at X,Y. Given $HA = 3$, $HX = 2$, $HY = 6$, write $[ABC] = m\sqrt{n}$ with n squarefree and find $m + n$.

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2020 AIME II Problem 2: A random slope in a square

Choose P uniformly in the unit square. Find numerator plus denominator of the chance that its slope from $(5/8, 3/8)$ is at least $1/2$.

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2020 AIME II Problem 4: A quarter turn with an unknown center

A clockwise rotation through m degrees, $0 < m < 180$, takes vertices $(0, 0), (0, 12), (16, 0)$ to $(24, 18)$, $(36, 18), (24, 2)$ respectively. If the center is (x, y), find $m + x + y$.

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2020 AIME II Problem 7: The largest sphere in two perpendicular cones

Two congruent solid right circular cones have radius 3 and height 8. Their perpendicular axes meet at an interior point O,3 units from each base. A sphere must lie inside both cones. If the greatest radius squared is m/n reduced, find m+n.

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2020 AIME II Problem 12: A line crossing a numbered grid

An odd m by odd n row-major grid has label 200 in its top row and 2000 in its bottom row. Count (m, n) for which the line through those cell centers enters cell 1099.

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2020 AIME II Problem 13: Tangent lengths and a monotone angle sum

A convex pentagon ABCDE has an incircle and side lengths $AB = 5$, $BC = CD = DE = 6$, $EA = 7$. Find its area.

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2020 AIME II Problem 15: A projection parallelogram from two circle tangents

In acute scalene triangle ABC, the tangents at B,C to its circumcircle meet at T. Points X,Y are the perpendicular projections of T onto lines AB,AC. Given $BT = CT = 16$, $BC = 22$, and $TX^2 + TY^2 + XY^2 = 1143$, find XY^2.

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2021 AIME I Problem 2: Two rectangles sharing a diagonal

Rectangles ABCD and AECF share diagonal AC. The first has $AB = 3$, $BC = 11$; the second has $AF = 7$, $FC = 9$. Their placement has E beyond side BC and F beyond side AD, so AE crosses BC and CF crosses AD. If the overlap area is m/n reduced, find m+n.

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2021 AIME I Problem 6: A spatial British-flag identity

AB,AC,AD are the three cube edges from A, and G is the opposite vertex. A point P has distances $BP = 60\sqrt{10}$, $CP = 60\sqrt{5}$, $DP = 120\sqrt{2}$, $GP = 36\sqrt{7}$. Find AP.

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2021 AIME I Problem 9: Three altitudes determine a trapezoid through a side ratio

An isosceles trapezoid ABCD has $AB \parallel CD$, $AD = BC$, and $AB < CD$. The distances from A to lines BC,CD,BD are 15,18,10. If its area is K, find $\sqrt{2}K$.

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2021 AIME I Problem 11: Projecting a quadrilateral onto its opposite diagonals

A cyclic quadrilateral has side lengths $AB = 4$, $BC = 5$, $CD = 6$, $DA = 7$. Project A,C orthogonally to line BD to obtain A_1, C_1, and project B,D to AC to obtain B_1, D_1. If the perimeter of $A_1B_1C_1D_1$ is m/n reduced, find m+n.

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2021 AIME I Problem 13: An affine power difference measures a radical-axis distance

Circles of radii 961 and 625 intersect at A,B. A third circle is externally tangent to both. Line AB cuts a 120-degree minor arc from the third circle. Find the distance between the centers of the first two circles.

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2021 AIME I Problem 15: Perpendicular parabolas share a circle of intersection

For positive integer k, consider $y = x^2 - k$ and $x = 2(y-20)^2 - k$. Let S contain k for which the curves meet at four distinct points on a circle of radius at most 21. Find the sum of the least and greatest members of S.

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2021 AIME II Problem 2: An equilateral slice and an exterior isosceles triangle

ABC is equilateral of side 840. Point D lies on the perpendicular to BC through B, toward A. The parallel to BC through D meets AB,AC at E,F. Extend EF beyond F to G so AFG is isosceles. If $[AFG] : [BED] = 8 : 9$, find AF.

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2021 AIME II Problem 5: When fixed sides and area specify one obtuse triangle

For s>0, consider all obtuse triangles of area s having sides 4 and 10. The values of s for which such triangles exist and are all congruent form $[a, b)$. Find $a^2 + b^2$.

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2021 AIME II Problem 8: Nonbacktracking walk on a cube

On a cube, an ant starts at a bottom vertex, chooses its first edge uniformly, and thereafter uniformly chooses either edge that does not reverse its previous step. Find numerator plus denominator of its chance of being on top after 8 steps.

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2021 AIME II Problem 10: Three spheres determine the opening of a common wedge

Three pairwise externally tangent spheres have radii 36,36,13. Two distinct common tangent planes meet in line ell. Find the distance from ell to the 13-sphere's tangency point on either plane, expressed as m/n reduced; report m+n.

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2021 AIME II Problem 12: Diagonal angle from area and alternating side squares

A convex quadrilateral has consecutive side lengths 5,6,9,7 and area 30. Let theta be its acute diagonal angle. Write $\tan\theta = m/n$ reduced and find m+n.

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2021 AIME II Problem 14: Two right-angle circles transfer an angle

An acute triangle ABC has circumcenter O and centroid G. The tangent at A meets the perpendicular to OG through G at X; line XG meets BC at Y. Angles ABC,BCA,XOY have ratio 13:2:17. Write angle BAC as m/n degrees reduced and find m+n.

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2022 AIME I Problem 3: Angle bisectors meet on the trapezoid midline

An isosceles trapezoid ABCD has parallel bases AB=500, CD=650 and legs AD=BC=333. The internal bisectors at A,D meet at P; those at B,C meet at Q. Find PQ.

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2022 AIME I Problem 5: Subtract squared swimming velocities

A 264-foot-wide river flows east at 14 feet per minute. Melanie, swimming at 80 feet per minute relative to the water, starts D feet downstream from Sherry, whose swimming speed is 60. They swim straight to the same opposite-bank point, equally far from their starting points, and arrive together. Find D.

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2022 AIME I Problem 8: Intersections of the three vertex circles

An equilateral triangle has circumradius 18. At each vertex, a circle is tangent to both incident side segments and internally tangent to the circumcircle. Their six pairwise intersections form two equilateral triangles. Write the smaller triangle's side as $\sqrt{a}-\sqrt{b}$, with positive integers a,b, and find a+b.

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2022 AIME I Problem 10: Equal-radius slices of tangent spheres

Three pairwise externally tangent spheres have radii 11,13,19. A plane cuts them in equal-radius circles centered A,B,C, respectively. All sphere centers lie on the same side of the plane. Given AB squared=560, find AC squared.

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2022 AIME I Problem 11: Tangent lengths determine a parallelogram

A circle is tangent to segments DA,AB,BC of parallelogram ABCD, whose angle A is acute. Diagonal AC meets the circle at P,Q in that order, with AP=3,PQ=9,QC=16. Write the parallelogram's area as $a\sqrt{b}$ with b squarefree; find a+b.

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2022 AIME I Problem 14: Perimeter-bisecting cleavers determine a120-degree angle

A triangle has BC=219 and positive integer sides AB,AC. The lines through the midpoints of AB and AC that divide the triangle into two regions of equal perimeter meet at 30 degrees. Find its perimeter.

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2022 AIME II Problem 3: Circumsphere of a right square pyramid

A right square pyramid has base side 6 and volume 54. Its five vertices lie on a sphere of radius m/n in lowest terms. Find m+n.

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2022 AIME II Problem 7: The triangle of common tangents

Two externally tangent circles have radii 6 and 24. Their three common tangent lines form a triangle. Find its area.

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2022 AIME II Problem 9: Regions between two supporting lines

Join all m marked points on one line to all n on a parallel line, with no triple interior intersection. Including the supporting lines, count bounded regions for $m = 7, n = 5$.

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2022 AIME II Problem 11: Two angle-bisector reflections determine a quadrilateral

A convex quadrilateral ABCD has AB=2,AD=7,CD=3. The acute angles DAB and ADC have internal bisectors meeting at the midpoint M of BC. Find the square of its area.

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2022 AIME II Problem 12: Pair the foci of two intersecting ellipses

Reals a>4,b>1,x,y satisfy $x^2/a^2 + y^2/(a^2-16) = 1$ and $(x-20)^2/(b^2-1) + (y-11)^2/b^2 = 1$. Find the minimum of a+b.

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2022 AIME II Problem 15: Ptolemy and opposite-angle products in a cyclic hexagon

Two externally tangent circles have centers U,V at distance 15. A third circle through U,V meets the first at B,C and the second at A,D, in convex cyclic order A,B,U,C,D,V. Given AB=2 and CD=16, find the hexagon area.

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2023 AIME I Problem 1: No opposite pair of men

Five men and nine women stand randomly at 14 cyclic positions. Find numerator plus denominator of the probability every man is opposite a woman.

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2023 AIME I Problem 5: Two diameter triangles recover a square's area

P lies on the circumcircle of square ABCD, with PA times PC=56 and PB times PD=90. Find the square area.

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2023 AIME I Problem 8: Use inward normals to settle an angle branch

A rhombus ABCD has acute angle A. P lies on its incircle and has distances 9,5,16 from lines DA,AB,BC, respectively. Find the perimeter.

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2023 AIME I Problem 12: Sum three directed projections in an equilateral triangle

In equilateral ABC of side 55, D,E,F lie on BC,CA,AB with BD=7,CE=30,AF=40. An interior P satisfies angle AEP=angle BFP=angle CDP=alpha. Find tan squared alpha.

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2023 AIME I Problem 13: A Gram determinant distinguishes two rhombohedra

Every face of each of two noncongruent parallelepipeds is a rhombus with diagonals $\sqrt{21}$ and $\sqrt{31}$. Write the larger volume divided by the smaller as m/n reduced; find m+n.

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2023 AIME II Problem 3: A Brocard point gives two similar triangles

In an isosceles right triangle ABC with right angle A, an interior point P satisfies angle PAB=angle PBC=angle PCA and AP=10. Find the triangle area.

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2023 AIME II Problem 6: Midpoints in an L-shaped region

Choose two independent uniform points in a 2 by 2 square with its upper-right unit square removed. Find numerator plus denominator of the chance their midpoint remains in the region.

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2023 AIME II Problem 7: Avoiding monochromatic rectangles

Color 12 cyclic vertices red or blue. Count colorings with no monochromatic rectangle among the vertices.

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2023 AIME II Problem 9: A common tangent and radical axis determine a trapezoid

Circles meet at P,Q with PQ=5. Their common external tangent nearer P touches at A,B. The parallel through P meets them again at X,Y, with PX=10,PY=14. Write the area of trapezoid XABY as m $\sqrt{n}$, n squarefree; find m+n.

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2023 AIME II Problem 12: Use the stated uniqueness to construct the point

Triangle ABC has AB=13,BC=14,CA=15, and M is the midpoint of BC. The ray AM meets the circumcircle again at P beyond M. A unique Q on AM satisfies angle PBQ=angle PCQ. Write AQ=m/$\sqrt{n}$, with m,n coprime positive integers; find m+n.

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2023 AIME II Problem 14: Cut a tilted cube into two simplexes

A cube has parallel edges AB,CD and face diagonals AC,BD, with rectangle ABDC in a vertical plane. Their heights above a horizontal plane are 0,2,8,10 meters, respectively. Water fills the cube up to height 7. Write its volume as m/n reduced; find m+n.

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2024 AIME I Problem 5: Power of a point between offset rectangles

Rectangles ABCD and EFGH lie on opposite sides of their common baseline, with D,E,C,F in that order. A,D,H,G are concyclic. Given BC=16,AB=107,FG=17,EF=184, find CE.

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2024 AIME I Problem 8: A row of congruent circles determines the inradius

Inside triangle ABC, a row of eight circles of radius 34 is tangent to BC, with consecutive circles tangent and the two end circles tangent to AB,AC. A row of 2024 radius-one circles fits in the same way. Write the inradius as m/n reduced; find m+n.

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2024 AIME I Problem 9: A rhombus approaches a hyperbola's asymptote

Four points on $x^2/20 - y^2/24 = 1$ form a rhombus with diagonal intersection at the origin. Find the greatest real number strictly below BD squared for every such rhombus.

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2024 AIME I Problem 10: Tangent similarities and Ptolemy determine a symmedian chord

Triangle ABC has AB=5,BC=9,CA=10. The circumcircle tangents at B,C meet at D. Segment AD meets the circle again at P. Write AP=m/n reduced; find m+n.

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2024 AIME I Problem 11: Rotating blue vertices away from themselves

Color the eight vertices of a regular octagon independently red or blue. Find numerator plus denominator of the probability that some rotation sends every blue vertex to a red vertex.

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2024 AIME I Problem 14: Place a disphenoid in a rectangular box

A tetrahedron has AB=CD=$\sqrt{41}$, AC=BD=$\sqrt{80}$, and BC=AD=$\sqrt{89}$. Its insphere radius is m $\sqrt{n}$/p in reduced squarefree form. Find m+n+p.

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2024 AIME I Problem 15: A double-root boundary gives the largest box diagonal

Consider all rectangular boxes of surface area 54 and volume 23. Let r be the least radius such that each box fits inside a sphere of radius r. Write r squared=p/q reduced; find p+q.

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2024 AIME II Problem 5: Similar corner triangles give a reciprocal-side formula

A convex equilateral hexagon has each pair of opposite sides parallel. The extended lines of alternating sides form a triangle with side lengths 200,240,300. Find the hexagon side length.

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2024 AIME II Problem 8: Axial sections reduce torus tangency to two circles

A torus has tube radius 3 and its tube centers are 6 units from its axis. A sphere has radius 11. When the torus is internally or externally tangent to the sphere along a circle, those circle radii are r_i,r_o. Write r_i−r_o=m/n reduced; find m+n.

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2024 AIME II Problem 10: A perpendicular center line forces a side-sum identity

Triangle ABC has circumcenter O, incenter I, circumradius 13, and inradius 6. If AI is perpendicular to OI, find AB times AC.

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2024 AIME II Problem 12: The unique ladder through a point touches an astroid

Let A=(1/2,0),B=(0,$\sqrt{3}$/2). Among unit segments joining the positive coordinate axes, there is a unique interior point C of AB lying on no other such segment. Write OC squared=p/q reduced, where O is the origin; find p+q.

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2024 AIME II Problem 15: Rectangles supported by dodecagon chords

Count rectangles whose four sides lie on sides or diagonals of a regular dodecagon. Rectangle corners need not be polygon vertices.

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2025 AIME I Problem 2: Affine coordinates preserve a reflected heptagon's area

On AB, points A,D,E,B occur with consecutive lengths 4,16,8; on AC, A,F,G,C occur with lengths 13,52,26. Reflect D about F to get M, and G about E to get N. If DEGF has area 288, find the area of heptagon AFNBCEM.

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2025 AIME I Problem 6: A tangential trapezoid determines its base sums

An isosceles trapezoid of area 72 has an incircle of radius 3. Its unequal parallel bases have lengths r,s. Find r squared+s squared.

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2025 AIME I Problem 8: A circle has exactly one point on each tangent

For real k, the equations $|25 + 20i - z| = 5$ and $|z - 4 - k| = |z - 3i - k|$ have exactly one complex solution z. Write the sum of all possible k as m/n reduced; find m+n.

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2025 AIME I Problem 9: Use reflection symmetry to construct the required intersection

Rotate the parabola y=x squared−4 by 60 degrees counterclockwise about the origin. Its unique fourth-quadrant intersection with the original has y-coordinate $(a-\sqrt{b})/c$, with a,c coprime positive integers. Find a+b+c.

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2025 AIME I Problem 12: Order chambers divide an equilateral simplex

In the plane x+y+z=75, the strict inequalities $x-yz<y-zx<z-xy$ describe three disjoint convex regions, exactly one of finite area. Write that finite area as a $\sqrt{b}$ with b squarefree; find a+b.

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2025 AIME I Problem 13: Expected regions from random chords

Two perpendicular diameters divide a disk into four quadrants. Add 25 independent chords, each choosing uniformly a pair of distinct quadrants and then uniform endpoints within those quadrants. Find the expected number of regions.

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2025 AIME I Problem 14: Two Ptolemy bounds share an equality point

A convex pentagon ABCDE has consecutive sides 14,7,24,13,26 and angles B=E=60 degrees. Find m+n+p if the least value of AX+BX+CX+DX+EX over planar points X is m+n $\sqrt{p}$, with p squarefree.

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2025 AIME II Problem 1: Recover one base and reuse its altitude

Collinear points A,B,C,D,E,F occur in order, with AC=26,BD=22,CE=31,DF=33,AF=73. Off-line point G has CG=40,DG=30. Find the area of BGE.

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2025 AIME II Problem 5: Read minor arcs from the medial triangle

Triangle ABC has angles A=84,B=60,C=36 degrees. D,E,F are the midpoints of BC,CA,AB. Their circumcircle meets BD,AE,AF again at G,H,J, in cyclic order G,D,E,H,J,F. Find arc DE+2 arc HJ+3 arc FG in degrees.

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2025 AIME II Problem 6: Equal triangle areas cancel the rectangle's product

A radius-six circle centered A is internally tangent at B to a radius-fifteen circle. BC is a diameter of the larger circle, and AD is perpendicular to BC with D on that circle. A rectangle EFGH is inscribed in the smaller circle, EF perpendicular to BC, with C nearer GH and D nearer FG. If DGF and CHG have equal areas, write the rectangle area m/n reduced; find m+n.

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2025 AIME II Problem 11: Equal-length perfect matchings on a polygon

Pair all vertices of a regular 24-gon so that the twelve resulting chords have the same length. Count the pairings.

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2025 AIME II Problem 12: Equal-area fan triangles alternate their radial lengths

A nonconvex 11-gon A 1,...,A 11 has perimeter 20. For i=2 through 10, triangle Ai,A 1,A(i+1) has area 1 and cosine of its angle at A 1 equal 12/13. Write A 1 A 2+A 1 A 11 as $(m\sqrt{n}-p)/q$ in normalized squarefree form; find m+n+p+q.

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2025 AIME II Problem 14: An equilateral triangle inside a right triangle controls area

Right triangle ABC has $\angle A = 90^\circ$ and $BC = 38$. Interior points K, L satisfy $AK = AL = BK = CL = KL = 14$. The area of $BKLC$ is $n\sqrt{3}$; find n.

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2026 AIME I Problem 3: A sphere fitting inside a hemisphere

A hemisphere of radius 200 has its flat face on a horizontal disk of radius 200. Let T be the possible tangency points on the disk for a sphere of radius 42 contained in the hemisphere. If $[T]/[\text{disk}] = p/q$ in lowest terms, find $p + q$.

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2026 AIME I Problem 5: Two opposite rotations reduce to one cosine

Let $AB = 1$. Rotate A counterclockwise about B through an acute angle θ to A', then rotate B clockwise about A' through θ to B'. Given $AB' = 4/3$, write $\cos\theta = m/n$ in lowest terms and find $m + n$.

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2026 AIME I Problem 10: Area of a triangle and its rotated interlacing

Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$. Rotate it about its circumcenter to $A'B'C'$ so that $A'C' \perp BC$ and A', B lie on opposite sides of $B'C'$. Find the nearest integer to the area of hexagon $AA'CC'BB'$.

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2026 AIME I Problem 12: Sphere radii form an affine function

In a plane, $AB = 6$, $AC = 4$, and $\angle BAC = 90^\circ$. Reflect the centroid of ABC across BC to obtain D. Spheres of radii $1, 2, 3, r$ touch the plane at A, B, C, D on the same side. All also touch a second plane on the same side of it. Find numerator plus denominator of r in lowest terms.

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2026 AIME I Problem 14: Closure controls an equiangular pentagon

An equiangular pentagon has side-length squares summing to 308 and diagonal-length squares summing to 800. Its perimeter squared is $m\sqrt{n}$, where m, n are positive integers and n is squarefree. Find $m+n$.

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2026 AIME I Problem 15: Five rectangular cell loops

A cell loop consists of the boundary cells of an axis-aligned rectangle with both dimensions at least two. Count partitions of a 10×10 board into five cell loops.

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2026 AIME II Problem 3: An indented pentagon gives one residue class

A simple nonconvex pentagon $ABCDE$ has interior angles $A = E = 90°$, $B = D = 45°$, with $DE < AB$, $AE = 20$, $BC = 14\sqrt{2}$. Vertices B, C, D lie on one side of AE. How many integer lengths $AB < 2026$ make its area a multiple of 16?

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2026 AIME II Problem 6: Normal distances from a point to a parabola

Find the sum of all radii of circles centered at $(4, 39)$ that are tangent to the parabola $2y = x^2 - 8x + 12$ at at least one point.

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2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle

In an isosceles triangle ABC, $AB = BC$ and I is the incenter. The perimeters of ABC and AIC have ratio $125 : 6$. All side lengths of both triangles are integers. Find the least possible AB.

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2026 AIME II Problem 10: An angle bisector and tangent secants

In triangle ABC, $AB = 200$, $AC = 225$, and AD bisects angle A, with D on BC. A circle through A is tangent to BC at D and meets AB, AC again at E, F on those segments. If AE, AF, BD, CD are positive integers, find the greatest possible BC. An alternate version asks for the sum of all possible values.

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2026 AIME II Problem 12: The centers of a tetrahedron with two unequal opposite edges

A tetrahedron has two faces with side lengths $5\sqrt{10}, 5\sqrt{10}, 10$ and two with $5\sqrt{10}, 5\sqrt{10}, 18$. Its circumcenter and incenter are S, R. Write $RS = m/n$ in lowest terms and find $m + n$.

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